# TS3000 series Robot Controller

# INSTRUCTION MANUAL
# CONVEYOR SYNCHRONIZATION FUNCTION

Rev. 2

March 30, 2021

**Notice**

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before installing and operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

Copyright 2020 by SHIBAURA MACHINE CO.,LTD..
All rights reserved.

No part of this document may be reproduced in any form without obtaining prior written permission from SHIBAURA MACHINE CO.,LTD..

The information contained in this manual is subject to change without prior notice to effect improvements.

- Preface -

This manual covers information on the conveyor synchronization function of the industrial robot controller TS3000 of SHIBAURA MACHINE, such as the operational overview, specifications, and instruction words.

Section 1 Overview
Gives the overview of the conveyor synchronization function.

Section 2 Specifications
Explains the basic specifications for the conveyor synchronization function.

Section 3 Parameters
Explains how to specify the parameters for using the conveyor synchronization function.

Section 4 Instruction Words
Explains the robot language for the conveyor synchronization function.

Section 5 Interface
Explains preparations and notes for using the conveyor synchronization function.

Section 6 Programming and Teaching Operation
Explains teaching operation procedures for using the conveyor synchronization function.

Section 7 Operation
Explains how to stop and reset the conveyor synchronization function.

Section 8 Troubleshooting

Please note that the contents of this manual may be revised without prior notice. The following instruction manuals are available: Overview Manual, Startup Manual, Operator's Manual, Robot Language Manual, Interface Manual, Communication Manual, Installation & Transport Manual, and Maintenance Manual. These instruction manuals should be used with the instruction manual "Conveyor Synchronization Function" as needed.

Contents

**Contents** ..... **4**
**Section 1 Overview** ..... **6**
**Section 2 Specifications** ..... **9**
**Section 3 Parameters** ..... **10**
3.1. ① Encoder rotating direction ..... 11
3.2. ② Encoder sampling period ..... 12
3.3. ③ Conveyor stop judgment time ..... 14
3.4. ④ Type of conveyor synchronization ..... 14
3.5. ⑤: Speed error compensation function switch ..... 15
3.6. ⑥: Speed error compensation velocity rate ..... 15
3.7. ⑦: Speed error compensation acceleration rate ..... 15
3.8. ⑧ Travel value per pulse of encoder ..... 16
3.9. ⑨ Rotation center position X of arc conveyor, and ⑩ Rotation center position Y of arc conveyor (World Coordinate System) ..... 17
3.10. ⑩ Limitation width of synchronization (inner), and ⑫ Limitation width of synchronization (outer) ..... 17
3.11. ⑬ - ⑮: Reserved areas ..... 17
**Section 4 Instruction Words** ..... **18**
CONV ..... 19
SYNC ..... 21
UNSYNC ..... 22
XIN ..... 23
CNVSTAT ..... 25
**Section 5 Interface** ..... **27**
5.1. Connection of Trigger Input Signal ..... 27
5.2. Circuit Configuration ..... 27
5.2.1. Contact and Transistor Specifications ..... 28
5.2.2. Examples of Connecting Conveyor Trigger Input Signal ..... 29
5.2.3. Specifications for Conveyor Trigger Input Signal Cable ..... 30
5.3. Connection of Conveyor Encoder ..... 30
5.3.1. Specifications for Incremental Encoder ..... 30
5.3.2. Specifications for Incremental Encoder Connector ..... 30
5.3.3. Connection between Incremental Encoder and Robot Controller ..... 31
5.3.4. Relationship between Encoder Cable Length and Number of Twisted-pair Power Cables ..... 32
5.3.5. Preparation of Cable ..... 33
5.3.6. Determination of Adequate Encoder Resolution ..... 34
**Section 6 Programming and Teaching Operation** ..... **35**
6.1. Notes on Programming ..... 35
6.2. Overview of Programming ..... 37
6.3. Setting of Conveyor Synchronization Parameters CONV (A,B,C,D) ..... 39
6.3.1. Determination of "C: Conveyor synchronization delay distance" ..... 39
6.3.2. Determination of "D: Conveyor acceleration/deceleration distance" ..... 40
6.3.3. Teaching of "A: Conveyor synchronization start position and B: Conveyor synchronization end position" ..... 41
6.4. Teaching Operation ..... 42
6.4.1. Teaching of Robot Waiting Position ..... 42
6.4.2. Teaching of Motion Positions in Conveyor Synchronization ..... 43
6.5. Readjustment/Confirmation of Conveyor Synchronization Parameters "CONV (A,B,C,D)" ..... 44
6.6. Program Operation Check ..... 47

6.7. Sample Programs ..... 48
6.7.1. Handling Operation ..... 48
6.7.2. Pick & Place Operation ..... 49
**Section 7 Operation ..... 50**
7.1. Stop Operation ..... 50
7.2. Reset Operation ..... 52
7.3. Conveyor Stop ..... 52
7.4. Conveyor Synchronization Utility (Utility → Conv) ..... 53
7.5. Counting of Total Number of Encoder Pulses ..... 54
**Section 8 Troubleshooting ..... 55**
8.1. Error Messages ..... 55
8.2. Troubleshooting ..... 56

# Section 1 Overview

This section gives an overview of the conveyor synchronization function.

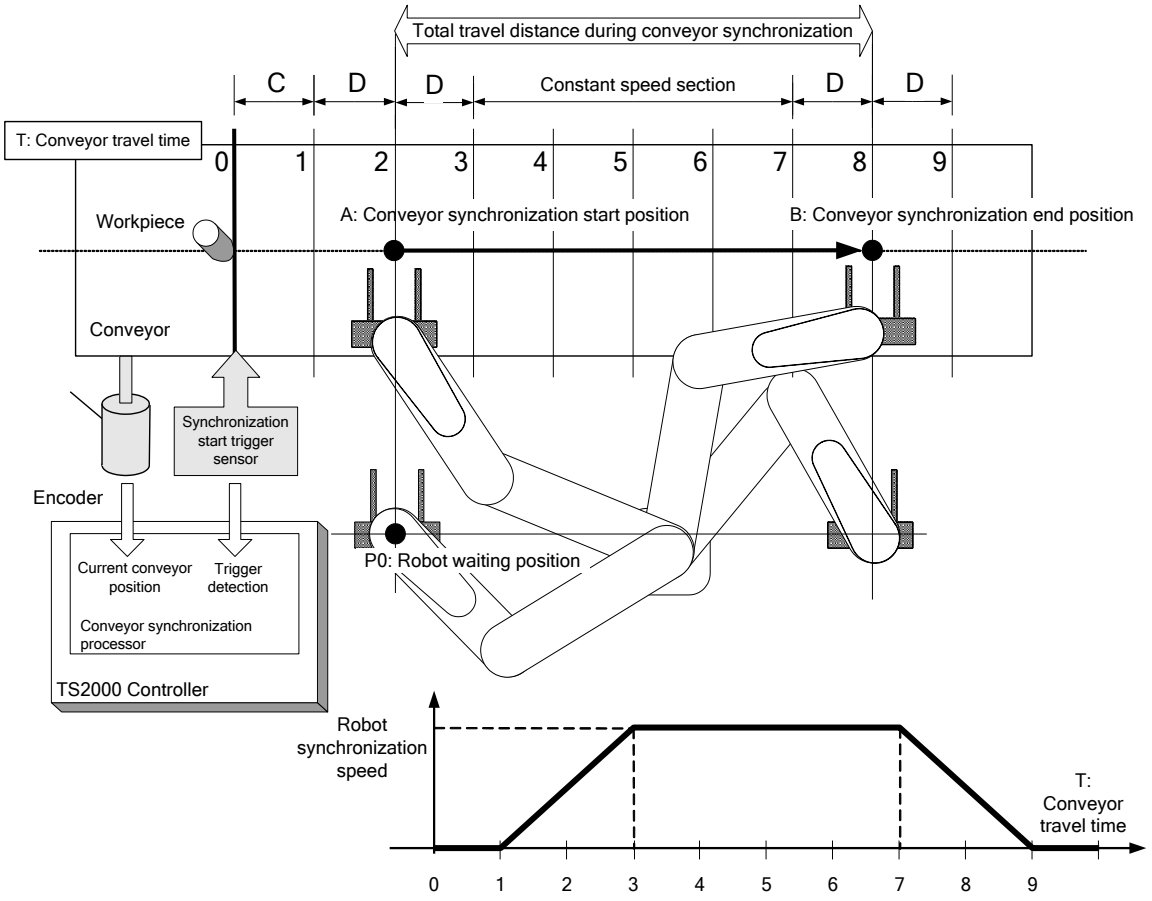



Figure 1.1 Schematic View of Conveyor Synchronization*

The conveyor synchronization processor reads the current conveyor position (speed) from the encoder mounted on the conveyor. At the moment when the trigger sensor detects a workpiece on the moving conveyor, the processor starts processing for the robot to catch up with the conveyor speed on the basis of the values of the encoder counter.

In order to specify the direction and travel distance of the conveyor to be synchronized, the conveyor synchronization processor also teaches the robot that the two points A and B in Figure 1.1 are parallel to the conveyor. As a result, the robot travels synchronously with the conveyor in the range of (B-A), in the straight-line direction from the point A to the point B. If a position to wait for conveyor synchronization is also necessary as shown in the above figure, "P0: Robot waiting position" is provided at a position aligned with the point A. **(Point A: Conveyor synchronization start position, Point B: Conveyor synchronization end position, P0: Robot waiting position)**

* C and D are variables independent of each other. The above figure is explained with C = D for simplification, but a different value can be specified each for C and D.

① **Conveyor travel time T "0-1": Delay section**
The robot stops at the waiting position from when the controller detects the trigger signal to when the conveyor travels the "C: Conveyor synchronization delay distance."

② **Conveyor travel time T "1-3": Acceleration section**
If the robot instantly catches up with the conveyor speed, a strong shock is applied to the robot and workpiece, or the robot vibrates. In order to avoid the problem, "D: Conveyor synchronization acceleration/deceleration distance" is to be specified for smoothly starting and stopping the robot. When T = 1, the robot starts accelerating at the point A, and catches up with the conveyor when T = 3. In other words, while the conveyor travels the distance of "2 Ds," the robot accelerates to the conveyor speed in the distance "D" from the point A.

③ **Conveyor travel time T "3-7": Constant speed section**
During conveyor synchronization (when the robot is travelling along with the conveyor), the robot travels parallel to the conveyor at the conveyor speed, so the robot does operation such as picking up workpieces during this time. (The motion program and operation positions during synchronization are taught to the robot in this section.)

④ **Conveyor travel time T "7-9": Deceleration section**
When T = 7, the robot starts decelerating, and comes to a complete stop at the point B when T = 9.
That is, while the conveyor travels the distance of "2 Ds," the robot decelerates to stop in the distance of "D."

## ⚠ CAUTION

1. Even if the program is stopped during conveyor synchronization, the robot can be continuously operated by immediately restarting the program. In principle, however, after operator intervention to perform a stop operation, be sure to perform a reset operation, move the robot to a safe place in manual guide mode, and then restart robot operation after resetting the program. (For details, see "7.2. Reset Operation.")

2. Unless an abnormal state is detected, do not perform any "stop operation with servo OFF" or "emergency stop operation" while the conveyor synchronization function is performed. Otherwise, the robot hand interferes with a workpiece on the conveyor in synchronization, which can lead to a collision accident.

3. To prevent the above accident, construct the system keeping the following in mind:

   ① If an external device (including a conveyor) is stopped in an emergency, the robot shall also be stopped in an emergency.

   ② If the robot is stopped in an emergency, the external device shall also be stopped in an emergency.

   ③ If the robot program is stopped for some reason (e.g. due to an error or a stop operation performed by the user, the conveyor shall also be stopped.

4. Before teaching operation can be performed, the conveyor must be stopped. If the synchronization start trigger sensor is activated during conveyor operation, an unexpected motion or collision can occur depending on the teaching operation condition. (The robot at the waiting position stays still only until the trigger sensor is turned on, but it is operable at any time.)

5. Define a safe place as a robot's evacuation position where the robot will never be involved in any collision with workpieces on the moving conveyor even if robot motion is stopped. (The robot waiting position is desired to be the evacuation position.)

6. Adjust the trigger sensor so that it will be always turned on at a fixed position without varying the detection position of objects (e.g. workpieces). If the sensor is turned on due to a mixture of different objects or a non-adjusted position, a follow-up delay or collision will occur.

7. In test run mode, do not execute a program (between the SYNC and UNSYNC instructions) in steps and do not use the machine lock (dry run) or MODE: MOTION. Otherwise, an unexpected collision can occur if the conveyor is operating.

# Section 2 Specifications

This section explains the specifications for the conveyor synchronization function.

Table 2.1 Basic Specifications for Conveyor Synchronization

| Item | Specifications | Remarks |
|---|---|---|
| 1. Number of axes*1 | Up to two axes can be supported for the conveyor. | The conveyor can be operated at a speed of 75-250 mm/sec. It can also be operated at a speed of 75 mm/sec or less, depending on the condition. |
| 2. Synchronization accuracy*2 | Accuracy: ±10 mm (with the conveyor speed 200 mm/sec)<br>• Positional interpolation sampling period: 3 msec<br>• Minimum interpolation unit (X, Y, Z): 0.001 mm<br>• Trigger signal sampling period: 500 μs | Accuracy shall be a value obtained when the conveyor speed variation is within ±7.5%. |
| 3. Teaching method | The robot is taught when the conveyor is in stop state. | |
| 4. Robot language | CONV, SYNC, UNSYNC, XIN, CONVSTAT | For details, see "Section 4 Instruction Words." |
| 5. Conveyor encoder interface*3 | Supply power: +5 V (internal supply)<br>System: 90 degree difference between the A and B phases<br>Specification: RS422 differential input (Line driver system)<br>Counter: 15-bit absolute value counter (0-32767)<br>Ports: 2 | Resolution:<br>Number of pulses per 3 msec<br>About "10-100 pulses" |
| 6. Trigger signal interface | Supply power: +24 V (external supply)<br>Specification: Photo coupler input<br>Ports: 2 | |

*1 If the conveyor is operated at a speed of 75 mm/sec or less, a high-resolution encoder must be used.

*2 The speed interpolation sampling period can be set by the parameter.

*3 The encoder resolution and the conveyor stop judgment time can be set by the respective parameters.

# Section 3 Parameters

The parameters for the conveyor synchronization function are to be specified in the "[C00] CONVEYOR PARAMETER" field of the conveyor parameter file "CONVEYOR.PAR." The parameter format is as shown below.

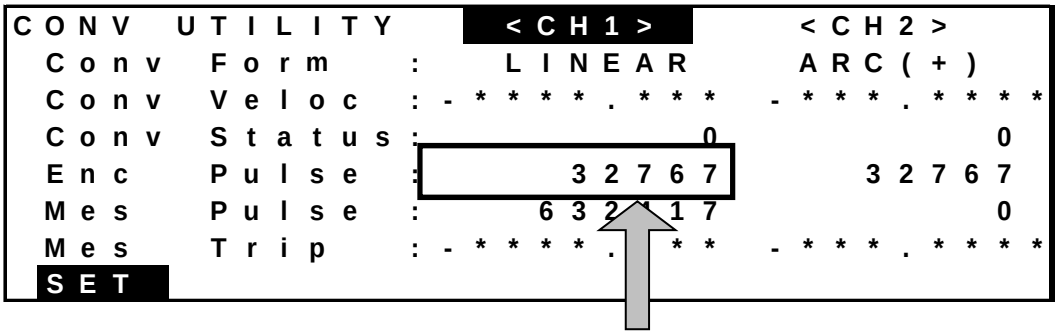
CONVEYOR.PAR

**[C00] CONVEYOR PARAMETER**
**{Conveyor Parameter1(CH1,CH2)}**
{Encoder rotating direction}
= 0 **0** ① Encoder rotating direction
{Encoder sampling period [ms]}
= 3 **3** ② Encoder sampling period [msec]
{Conveyor stop judgment time [msec]}
= 0 **0** ③ Conveyor stop judgment time [msec]
{Type of conveyor synchronization}
= 3 **3** ④ Type of conveyor synchronization function
{Speed error compensation function switch}
= 0 0 ⑤ Speed error compensation function switch
{Speed error compensation velocity rate}
= 0 0 ⑥ Speed error compensation velocity rate
{Speed error compensation acceleration rate}
= 0 0 ⑦ Speed error compensation acceleration rate
**{Conveyor Parameter2(CH1,CH2)}**
{Travel value per pulse of encoder [um/pulse] [mdeg/pulse]}
= 0.0 **0.0** ⑧Travel distance per encoder pulse [μm/pulse]
{Rotation center position X of arc conveyor [mm]}
= 0.0 **0.0** ⑨X-coordinate at the rotation center of the arc conveyor [mm]
{Rotation center position Y of arc conveyor [mm]}
= 0.0 **0.0** ⑩Y-coordinate at the rotation center of the arc conveyor [mm]
{Limitation width of synchronization(inner)}
= 0.0 **0.0** ⑪Area limited for conveyor synchronization (inner periphery) [mm]
{Limitation width of synchronization(outer)}
= 0.0 **0.0** ⑫Area limited for conveyor synchronization (outer periphery) [mm]
{Dummy data}
= 0 0 ⑬Self-explaining
{Dummy data}
= 0 0 ⑭Self-explaining
{Dummy data}
= 0 0 ⑮Self-explaining

### 3.1. ① Encoder rotating direction

Data format: Integer type
Data range: 0 or 1
Example value: 0

If the A-phase rising order and the B-phase rising order are compared when A-phase and B-phase pulses are output by the encoder, specify the addition direction to the A-phase fundamentally. The addition direction is determined by the specifications for the encoder and the rotating direction of the encoder mounted on the conveyor. After pressing the [Utility] key, and select the menu item "CONV." The menu screen as shown below will appear.




**Operate the conveyor, and set the parameter with the direction in which the value of the "Pulse" display in the figure increases.**

Examples:

① When the setting value for the encoder rotating direction is 0 and the displayed "Pulse" value increases: Set "0."

② When the setting value for the encoder rotating direction is 0 and the displayed "Pulse" value decreases: Set "1."

### 3.2. ② Encoder sampling period

Data format: Integer type
Data unit: msec
Data range: 3-30000
Example value: 3

Set the period in which the counter value of the conveyor encoder is to be obtained.

* The conveyor synchronization function performs a calculation every 3 msec, so a factor rounded down to the calculated period of "3 msec" is specified for this parameter.

<Note>
The conveyor synchronization function calculates the distance of synchronous travel with the minimum period of 3 msec by default. However, a value greater than the default value can be set. In this case, adjust the setting value, keeping the following in mind:

■ If a small value (minimum: 3 msec) is set:

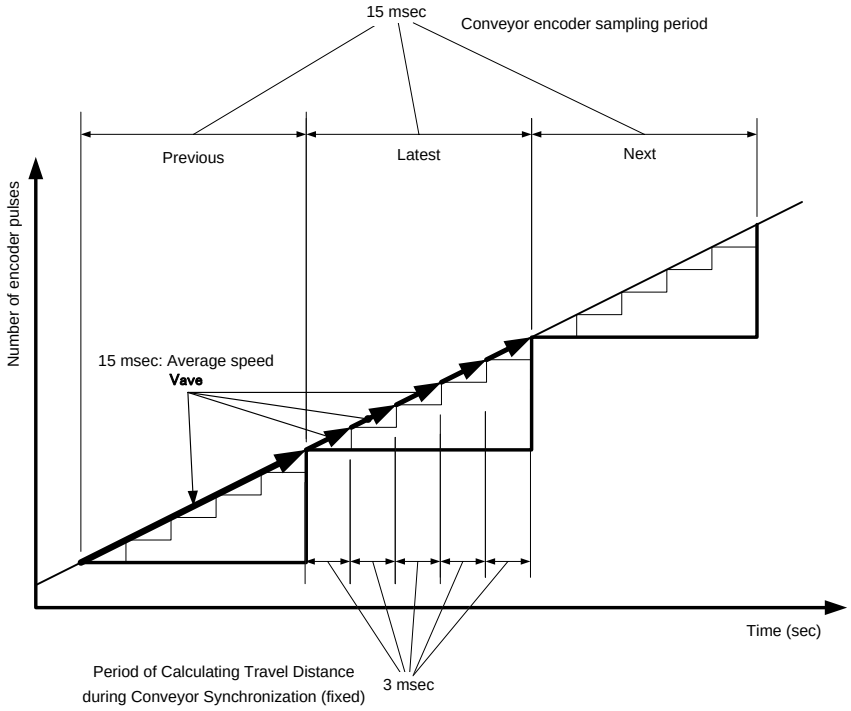
- Follow-up performance for the conveyor speed is improved.
- If too small a value is set, the robot can vibrate because it becomes sensitive when following the conveyor, responding even to conveyor pulsation.

■ If a large value is set:

- The robot can be made insusceptible to conveyor pulsation.
- An error in the calculation of the conveyor rotating angle can be absorbed.
- Follow-up performance for the conveyor speed is lowered.

If there is no conveyor pulsation, specify the default value (3 msec).

(Example)
If the conveyor sampling period is set to 15 msec, the conveyor synchronization function calculates the distance of synchronous travel on the basis of the average speed "Vave" in the previous conveyor sampling period.


15 msec
Conveyor encoder sampling period
Previous
Latest
Next
Number of encoder pulses
15 msec: Average speed
Vave
Time (sec)
Period of Calculating Travel Distance
during Conveyor Synchronization (fixed)
3 msec

### 3.3. ③ Conveyor stop judgment time

Specify the conveyor stop judgment time. If the encoder counter value does not change during the specified time, the system determines that the conveyor is in stop state, and issues the error message "001-048: Conveyor Stop." This error is a level-1 error, so the program runs continuously without being stopped.

Data format: Integer type
Data unit: msec
Data range: 0-30000
Example value: 3000

- The stop time is usually specified with the second argument of the SCOL-language "SYNC" instruction. If the argument is omitted, the system determines whether the conveyor is in stop state, depending on the setting value of the conveyor stop judgment time parameter.

- If the second argument of the "SYNC" instruction is omitted and the parameter setting is "0," the system does not judge whether the conveyor is in stop state.

### 3.4. ④ Type of conveyor synchronization

Data format: Integer type
Data range: 0-2
Example value: 0

Specify the type of the conveyor synchronization function, which can be selected from the following three:

0: Linear conveyor
1: Arc conveyor (clockwise)
2: Arc conveyor (counterclockwise)

Specify "0" for this parameter since a liner conveyor is to be used.

### 3.5. ⑤: Speed error compensation function switch

Enable/Disable the conveyor synchronous speed error compensation function.

Data format: Integer type
Data range: 0 or 1
0: Disable
1: Enable

### 3.6. ⑥: Speed error compensation velocity rate

This parameter is valid only for 6-axis systems.
Set the correction gain when the position error cannot be fully corrected to the conveyor constant speed.

Data format: Integer type

### 3.7. ⑦: Speed error compensation acceleration rate

This parameter is valid only for 6-axis systems.
Set the correction gain when the position error cannot be fully corrected during conveyor acceleration or deceleration.

Data format: Integer type

### 3.8. ⑧ Travel value per pulse of encoder

Obtain the distance (μm) of conveyor travel per pulse of the conveyor encoder.

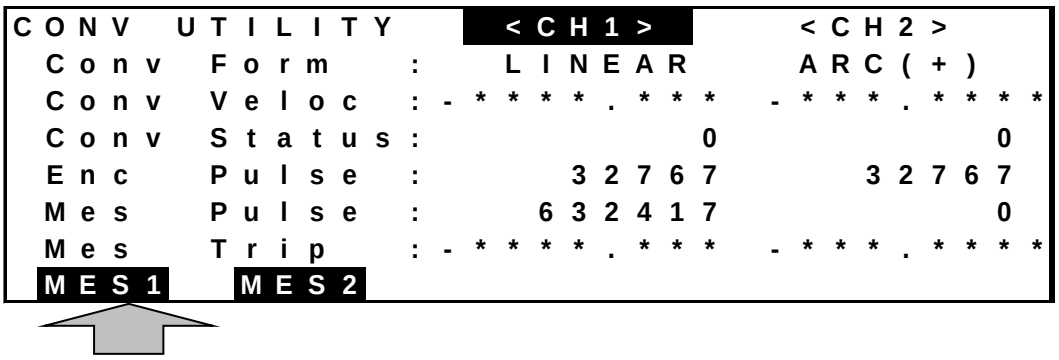
Data format: Real type
Data unit: μm
Example value: 0

If the total number of encoder pulses is assumed as ΔP (pulse) when the conveyor is moved by ΔL (mm), the distance of conveyor travel α (μm) per encoder pulse is obtained from the following equation:

$$\alpha = ( \Delta L / \Delta P )*1000$$

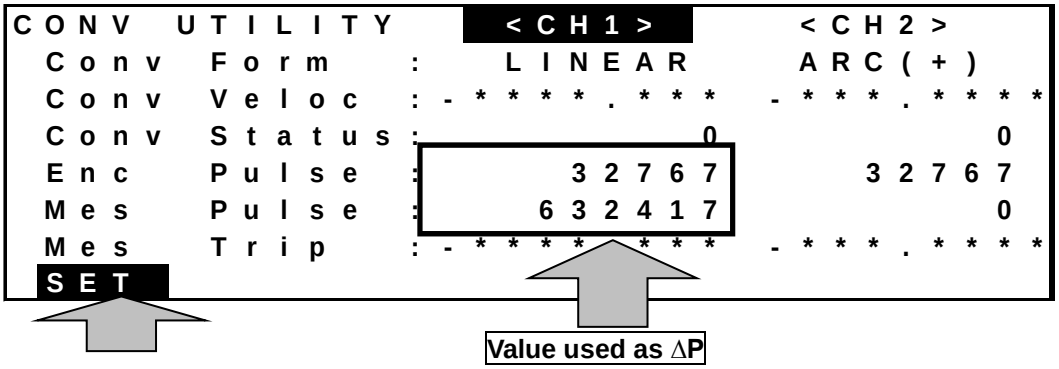
Shown below is the procedure to obtain the total number of encoder pulses ΔP.

① Using the cursor, select and press the function key <F1: MES1> or <F2: MES2> corresponding to the ch of which pulses are counted.




Select MES1 when counting the number of pulses of Ch1.

② Select "MES1" to count the pulses of the Ch1; select "MES2" to count the pulses of the Ch2. Move the conveyor the distance ΔL, and then press <SET>.




③ When <SET> is pressed, the current total number of pulses "Mes Pulse" is displayed on the screen. Assuming that the value is "ΔP," assign it to the above equation (4.1) and obtain the distance of conveyor travel α (μm) per encoder pulse. (To count pulses again, press MES1 or MES2.)

### 3.9. ⑨⑩ Rotation center position X of arc conveyor, and Rotation center position Y of arc conveyor (World Coordinate System)

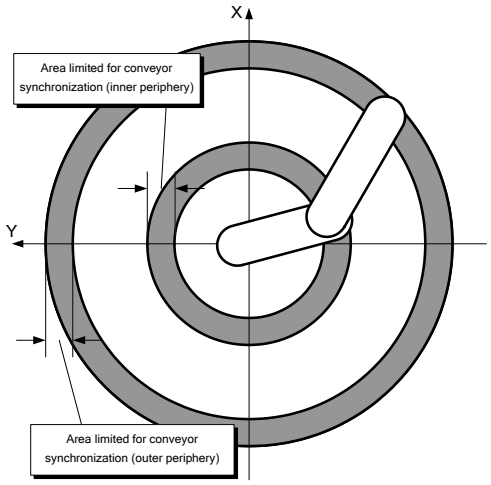
The parameters are to be used with the "arc conveyor synchronization function."

### 3.10. ⑪⑫ Limitation width of synchronization (inner), and Limitation width of synchronization (outer)

Data format: Real type
Data unit: mm
Data range: 0.0 or greater
Example value: 5.0 10.0

If the robot enters an area defined as "limitation width of synchronization" during conveyor synchronization, the system issues the error message "004-051: Conveyor sync limit area," and the robot decelerates to stop.
In this condition, conveyor synchronization is also stopped.

The parameters are to be specified with both the distance from the inner circle where robot motion is limited and the distance from the outer circle where robot motion is limited.




CONVEYOR.PAR: [C00] CONVEYOR PARAMETER
:
**{Limitation width of synchronization(inner)(mm)}**
5.0 10.0 ← Distance from the inner circle where robot motion is limited (mm)
**{Limitation width of synchronization(outer)(mm)}**
5.0 10.0 ← Distance from the inner circle where robot motion is limited (mm)

### 3.11. ⑬ - ⑮: Reserved areas

Data format: Real type

* Areas reserved for the system

# Section 4 Instruction Words

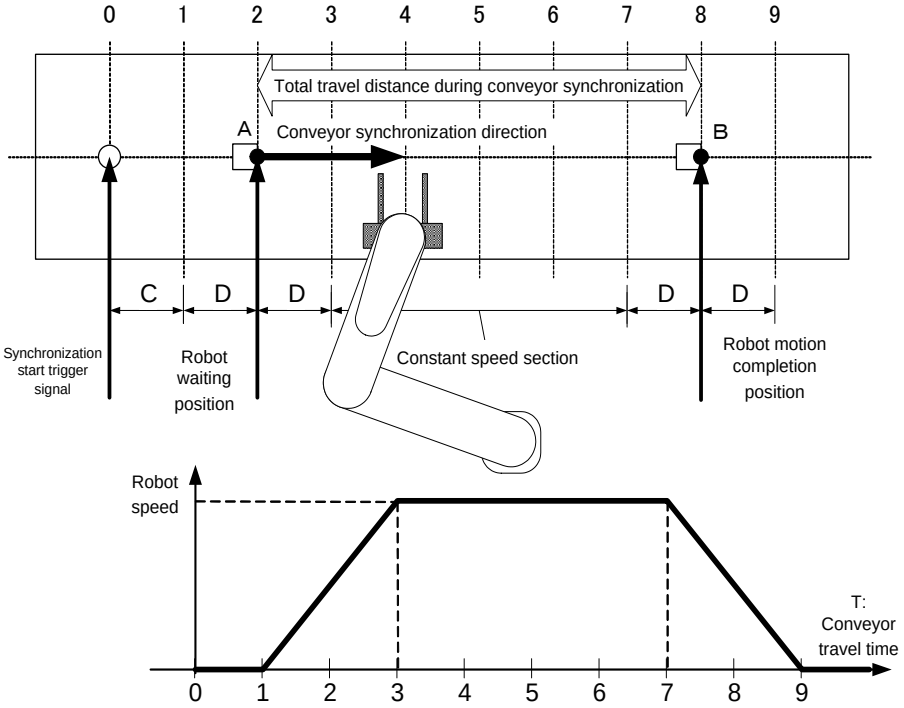
This section explains the "instruction words" for the conveyor synchronization function.

The following five instruction words are to be used as the language of SCOL (Symbolic Code Language for robot) for the conveyor synchronization function.

| Instruction word | Function | Page |
|---|---|---|
| CONV | Defines parameters for conveyor synchronization. | 4-2 |
| SYNC | Starts conveyor synchronization. | 4-4 |
| UNSYNC | Ends conveyor synchronization. | 4-5 |
| XIN | Monitors the encoder cable for a breakage. | 4-6 |
| CNVSTAT | Refers to conveyor synchronization status information. | 4-8 |

# CONV

| | |
|---|---|
| Function | : Defines parameters for conveyor synchronization. |
| Format | : CONV(<number>, <position 1>, <position 2>, <relative-amount 1>, <relative-amount 2>, <relative-amount 3>) |
| Sample statement | : CONV (N, A, B, C, D, E) |
| Comments or notes | : The parameters necessary for conveyor synchronization are to be defined. |




<number;N> **Logical conveyor number**
Specify the logical number of the conveyor to be synchronized. One (1) or two (2) can be specified. (Constants and variables can be used with scalar-type data.)

<position-1;A> **Conveyor synchronization start position** (position-type data)
Specify the position where conveyor synchronization is to start. At this position, the robot starts traveling synchronously with the conveyor. (Only position-type data can be specified with vector-type data.)

<position-2;B> **Conveyor synchronization end position** (position-type data)
Specify the position where synchronization between the conveyor and the robot is to end. At this position, the robot stops traveling synchronously with the conveyor. (Only position-type data can be specified with vector-type data.)
Assuming that $A=(X_A,Y_A,Z_A)$, $B=(X_B,Y_B,Z_B)$ , the travel distance during conveyor synchronization is obtained as follows:

$$|\overrightarrow{AB}| = \sqrt{(X_B-X_A)^2+(Y_B-Y_A)^2+(Z_B-Z_A)^2}\ 。$$

<relative-amount-1;C> **Conveyor synchronization delay distance (mm)**

• Specify the relative distance between the synchronization start trigger signal position with consideration of servo response delay and the robot waiting position. This parameter is to be specified in order to compensate for a physical synchronization delay. The specification of a longer distance than necessary can cause an error in the distance itself. In order to avoid this problem, the system should be constructed so that the parameter can be specified with the shortest distance for just fine adjustment.

<relative-amount-2;D> **Conveyor synchronization acceleration/deceleration distance (mm)**

Specify the acceleration/deceleration distance (unit: mm) necessary for conveyor synchronization.

If the condition of $D \leq \frac{|\overrightarrow{AB}|}{2}$ is not satisfied, the system issues the error message "002-033:
Conveyor Parameter Error," with which the robot stops. Conveyor synchronization is complete when the conveyor travels the distance between the points A and B after the synchronization is started.

<relative-amount-3;E> **Travel distance per encoder pulse (μm)**

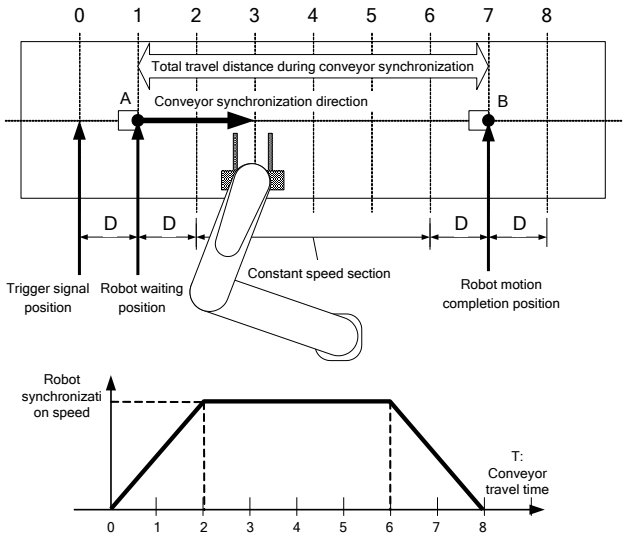
This parameter is identical to [C00] CONVEYOR PARAMETER (Travel value per pulse of encoder). See "3.6. ® Travel value per pulse of encoder" for how to obtain the parameter value. The E parameter can be omitted.

- When the E parameter is omitted or when E = 0: The value of the conveyor parameter is used.
- When the conveyor parameter = 0: The value of CONV( ,,,, E) is used.

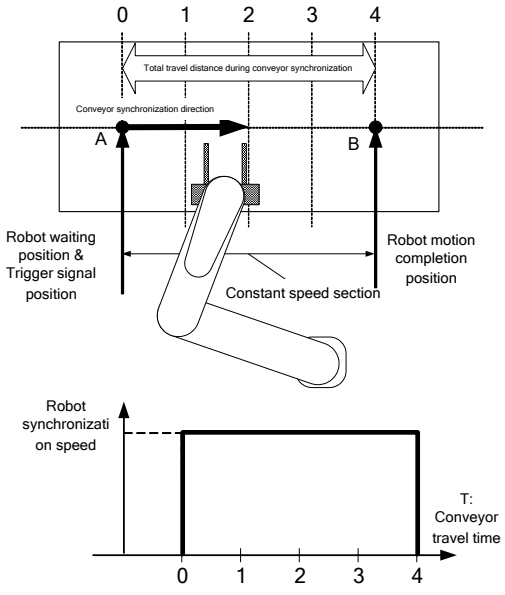
<Note>
When the conveyor parameter = 0.0 and the E parameter is omitted (or E = 0), the system issues the error message "002-033: Conveyor Parameter Error," with which the robot stops.

* Notes

- Once a conveyor synchronization parameter is defined, it is internally stored until the controller is turned off. To change the conveyor synchronization parameter setting, be sure to write a new CONV instruction and to perform a reset operation.
- Zero (0) can be specified for both the C and D parameters. However, when D (acceleration/deceleration distance) = 0, the robot rapidly increases the synchronization speed to the conveyor speed as shown in the following figure, in which case excess load can be applied to the robot.




When C = 0:




When C = D = 0:

# SYNC

| | |
|---|---|
| Function | : Starts conveyor synchronization. |
| Format | : SYNC(<number>,[<stop-judgment-time>]) |
| Sample statement | :SYNC(N, T) |
| Comments or notes | : Conveyor synchronization is started simultaneously when the synchronization start trigger signal is turned on. |

<number;N> Specify the logical number of the conveyor witch with synchronization is to be started. One (1) or two (2) can be specified, and constants and variables can be used with scalar-type data.

<stop-judgment-time> Specify the conveyor stop judgment time. If the item is omitted, the system determines whether the conveyor is in stop state, depending on the setting the value of [C00] CONVEYOR PARAMETER "Conveyor stop judgment time."

- If no conveyor synchronization parameter is defined in the CONV instruction, an error is issued when the SYNC instruction is performed.
- Be sure to specify the "SYNC" instruction when starting conveyor synchronization, and perform the UNSYNC instruction (explained later) at the position where the synchronization is complete. Otherwise, an error is issued when the SYNC instruction is performed.
- Keep in mind that the SYNC and UNSYNC instructions cannot be nested.
- If a stop operation is performed when the trigger signal is awaited to be turned on, the program stops without conveyor synchronization started. Before the program can be run again, the reset operation must be performed irrespective of the execution state (reset/retry/continue). For details, see "Section 7 Operation."

# UNSYNC

| | |
|---|---|
| Function | : Ends conveyor synchronization. |
| Format | : UNSYNC(<number>) |
| Sample statement | : UNSYNC(N) |
| Comments or notes | : Conveyor synchronization is ended. |

- <number;N> Specify the logical number of the conveyor with which synchronization is to be ended.
- If "<relative-amount-2;D> Conveyor synchronization acceleration/deceleration distance" specified in the CONV instruction is a non-zero value, the robot decelerates to stop during the time from when the UNSYNC instruction is performed to when the conveyor travels the distance "D."
- If synchronization between the points A and B "travel distance during conveyor synchronization: $|\overrightarrow{AB}|$" is complete, which is explained in the paragraph of the CONV instruction, the conveyor synchronization function is automatically terminated irrespective of whether or not the "UNSYNC" instruction is specified. (This does not mean that the UNSYNC instruction is not necessary, but means that the conveyor in synchronization is stopped in order to ensure safety.)
- If the UNSYNC instruction is not used paired with the SYNC instruction, an error is issued when the UNSYNC instruction is performed.

# XIN

| | |
|---|---|
| Function | : Monitors the encoder cable for a breakage. |
| Format | : XIN(<signal-name>) |
| Sample statement | : ON XIN(1) DO GOTO L100 |
| Comments or notes | : The encoder cable is monitored for a breakage, and an abnormality is informed if detected. |

<number;N> Specify the logical number of the conveyor whose encoder cable is to be monitored for a breakage.

- The XIN instruction is used for immediately detecting a breakage in the encoder cable. Specify the XIN instruction together with the ON instruction in the program so that the encoder cable can be always monitored for a breakage.
- Up to 10 pairs of XIN and ON-DO instructions can specified for always monitoring the encoder cable. Be careful not to exceed 10 pairs. If more than 10 pairs of these instructions are used, an error is issued, with which the robot program stops.
- Perform the IGNORE instruction to reset the encoder cable breakage monitoring. The IGNORE instruction can be used with the IF instruction and WAIT instruction.
- If a breakage in the encoder cable is detected, the system issues the error message "001-047 Conveyor Line Breakage." However, the robot does not stop, and it continues traveling synchronously with the conveyor at the conveyor (encoder) speed effective immediately before the encoder cable is broken.
- The error "001-047 Conveyor Line Breakage" can be reset by only turning off the power. Thus, if this error occurs, turn off the power, identify the location of the cable breakage, repair the encoder cable, and then turn the power back on.
- If a breakage in the encoder cable is detected during conveyor synchronization, be sure to move the robot to a safe place. Otherwise, the robot can collide with the workpiece or the conveyor. As shown in the sample program in the following page, if a cable breakage is detected, move the robot to a safe place by performing the MOVES instruction , end the conveyor synchronization by performing the UNSYNC instruction, and then stop the program by performing the STOP instruction.

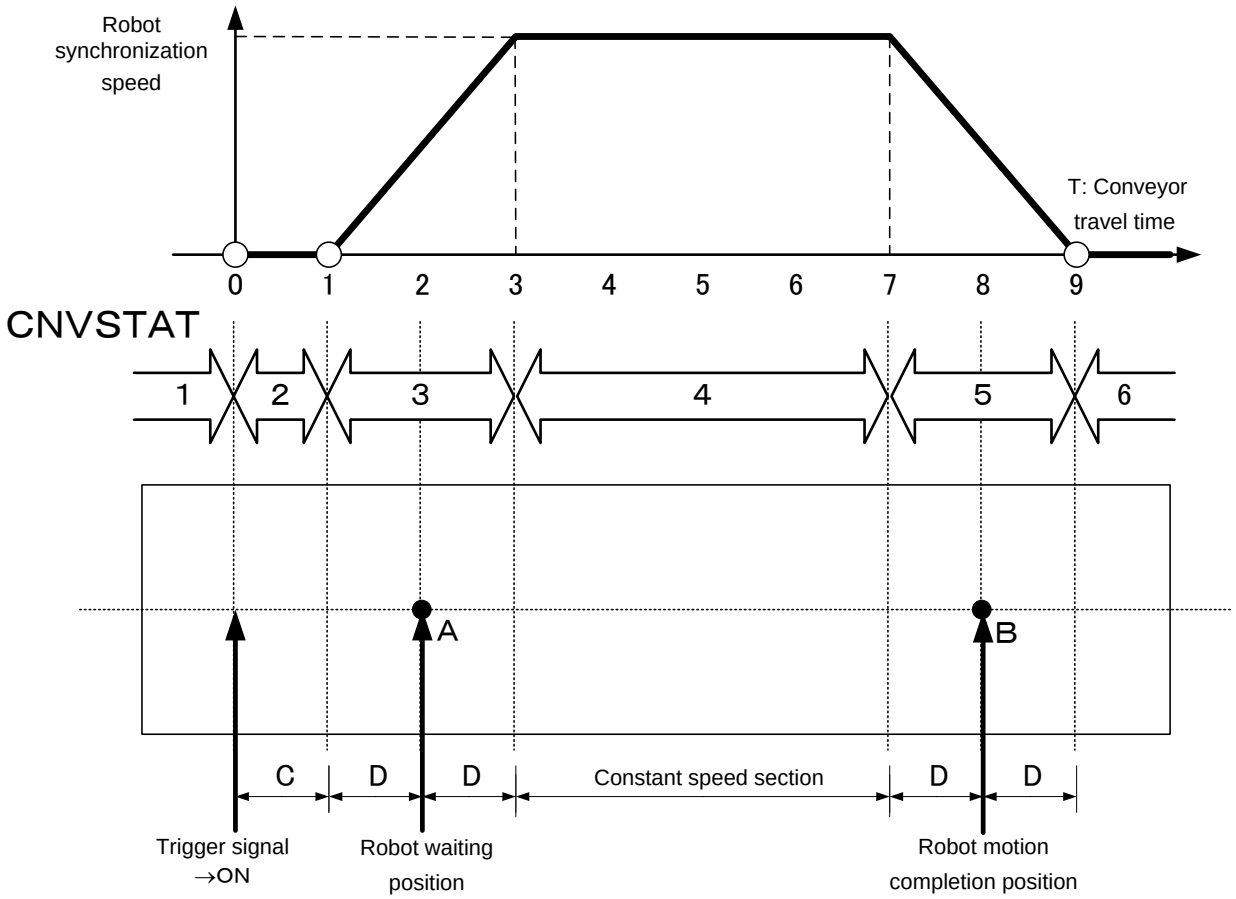
Sample Program

```
PROGRAM XINSAMPLE
        CONV( 1, A, B, 40.0, 10.0 )
        ••••••••••
L000:
        ON XIN(1) DO GOTO L100  The encoder cable is monitored for a breakage. - Before start of synchronization
        ••••••••••  (Monitoring must be reset immediately after the XIN instruction, so a cable breakage
                    check cannot be done.)
        IGNORE XIN(1)   Monitoring is reset. - Before start of synchronization
        ••••••••••
        SYNC(1)
        ON XIN(2) DO GOTO L200 The encoder cable is monitored for a breakage. - During synchronization
        ••••••••••
        IGNORE XIN(2)   Monitoring is reset. - During synchronization
        UNSYNC(1)

        GOTO L000
L100:
        PRINT "C/V ERROR(1)",CR
        STOP
L200:
        MOVES P1   After the robot is moved to a safe place by performing the linear interpolation control
                   instruction, the program is stopped.
        UNSYNC(N)
        PRINT "C/V ERROR(2)",CR
        STOP
END
```

# CNVSTAT

| | |
|---|---|
| Function | : Refers to conveyor synchronization status information. |
| Format | : CNVSTAT |
| Sample statement | : WAIT CNVSTAT > = 4<br>: ON CNVSTAT<>4 BREAK DO GOTO L1 |
| Comments or notes | : The conveyor synchronization status information is referred to. |



Conveyor Status Information

| | | |
|---|---|---|
| 0 | Conveyor in asynchronous state | Indicates that the conveyor is in asynchronous state. (The CNVSTAT instruction can be used only after the SYNC instruction is executed. Thus, "1" is never specified in the program.) |
| 1 | Trigger signal wait state | Indicates that the conveyor is waiting until the trigger signal is turned on. |
| 2 | Conveyor travel in delay section | Indicates that the conveyor is traveling in the delay section. |
| 3 | Conveyor travel in acceleration section | Indicates that the conveyor is traveling in the acceleration section. The robot catches up with the conveyor speed in this section. |
| 4 | Conveyor travel in constant speed section | Indicates that the conveyor is traveling in the constant speed section.<br>The robot is traveling, following the conveyor.<br>The workpiece on the conveyor must be handled while CNVSTAT is "4." |
| 5 | Conveyor travel in deceleration section | Indicates that the conveyor is traveling in the deceleration section. The robot catches up with the conveyor speed in this section. |
| 6 | Conveyor synchronization complete | Indicates that conveyor synchronization is complete and UNSYNC is waiting. |

* Note

- The CNVSTAT instruction can be used only during conveyor synchronization (i.e. after execution of the "SYNC" instruction).
- No value can be assigned to the CNVSTAT instruction.
- BREAK and PAUSE can be used only when the CNVSTAT conditional statement is used as the ON condition.

Sample Program

```
PROGRAM MAIN
    CONV(1,A,B,100.0,200.0)
    MOVE P1
    SYNC(1)
    WAIT CNVSTAT>=4
    MOVES P2
    MOVES P3
    WAIT CNVSTAT>=6
    UNSYNC(1)
END
```

# Section 5 Interface

## 5.1. Connection of Trigger Input Signal

- In order to detect the conveyor encoder position with higher accuracy when the trigger signal is turned on, the conveyor synchronization function detects the trigger signal by means of high-speed signal input.
- The conveyor synchronization start trigger uses two of the eight input points for the high-speed input function. For the CH1 conveyor, connect the trigger input signal to the high-speed signal number D149; for the CH2 conveyor, connect the trigger input signal to the high-speed signal number D150. (For details, see Figures 5.1 and 5.2.)
- If the high-speed input function is used with the conveyor synchronization start trigger switch, the switches of the six points other than the above two points are disabled. (The "position data latch function" using eight high-speed input points and the "conveyor synchronization function" cannot be used at the same time.)

Table 5.1 shows the high-speed signal input specifications for the TS3000.

Table 5.1 High-speed Signal Input Specifications for the TS3000

| Item | Specifications |
|---|---|
| Number of input points | Two input points (Only 2 of 8 points in total are used for conveyor synchronization.) |
| Input specifications | Transistor open-collector input, or no voltage contact input |
| Input signal (See Figure 5.1.) | ch1: DI49<br>ch2: DI50 |
| Delay time | 2.2 µsec (calculated value) |

## 5.2. Circuit Configuration

The circuit configuration is as shown in Figure 5.1. A bidirectional photo coupler is used with the input circuit, so SourceType (positive common) or SinkType (negative common) can be selected.

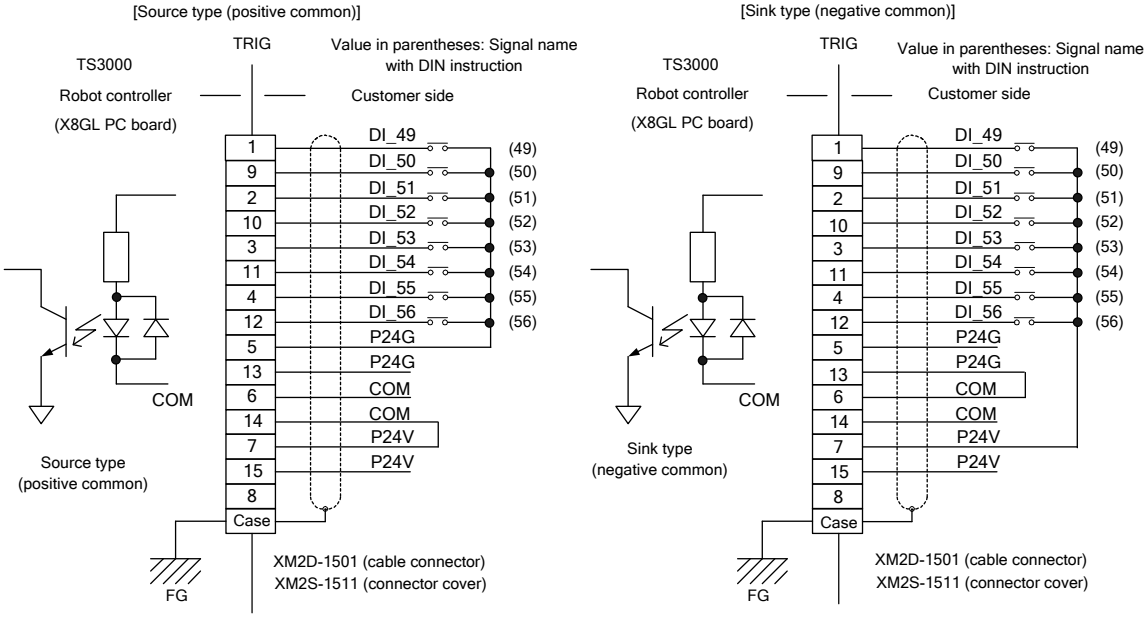



Figure 5.1 High-speed Input Circuit Connection Diagram

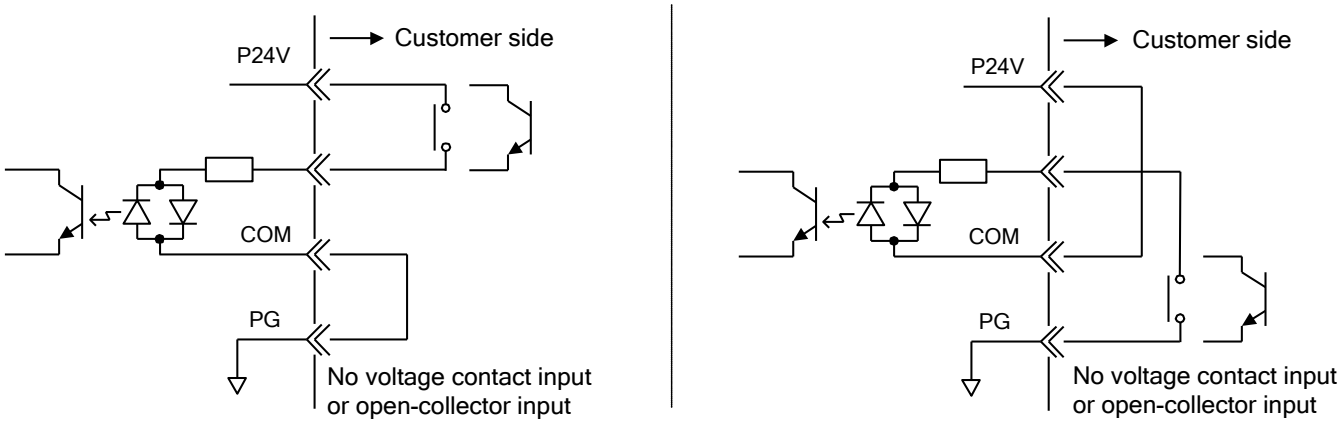



Figure 5.2 Examples of Input Signal Circuits (with internal power supply used)

### 5.2.1. Contact and Transistor Specifications

Only contacts and transistors that conform to the specifications shown in Table 5.2 can be used, connected externally.

Table 5.2 No Voltage Contact and Transistor Specifications

| No Voltage Contact Specifications | | Transistor Specifications | |
|---|---|---|---|
| Contact rating: | 24 VDC 10 mA or more | Collector-emitter voltage: | 30 V or more |
| Circuit current: | Approx. 7 mA | Collector-emitter current: | 10 mA |
| Minimum current: | 24 VDC 10 mA | Circuit current: | Approx. 7 mA |
| Connection impedance: | 100Ω or less | Collector-emitter leak current: | 100 μA |

### 5.2.2. Examples of Connecting Conveyor Trigger Input Signal

Connect the conveyor synchronization start trigger switch to either of the conveyor synchronization trigger switches 1 and 2 for high-speed input. Figures 5.3 and 5.4 show an example of connecting the conveyor synchronization start trigger switch.

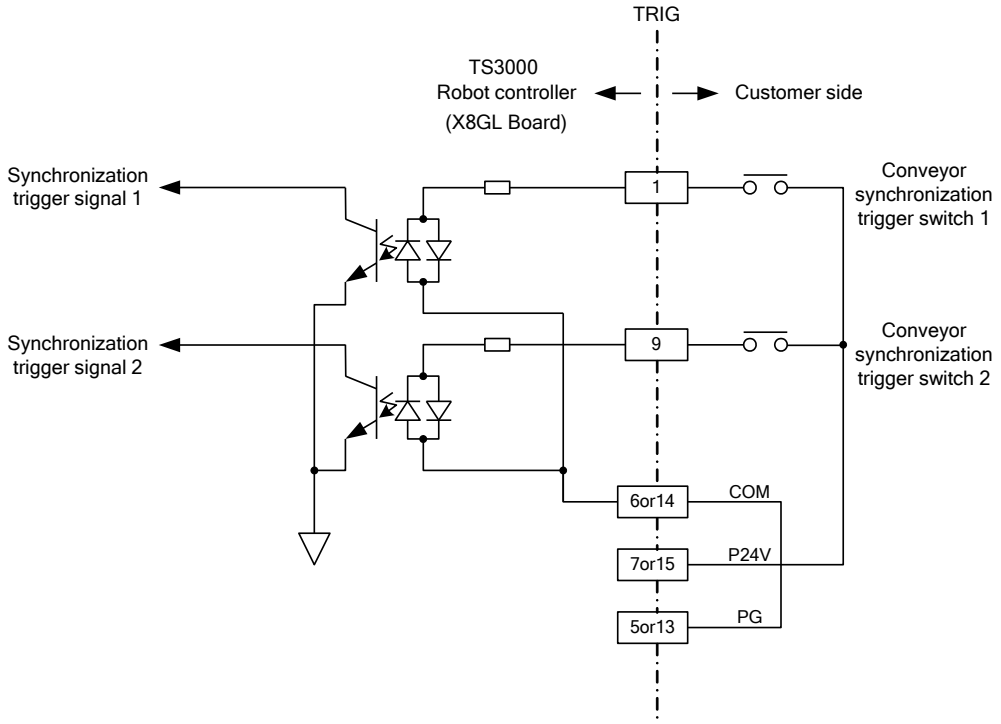
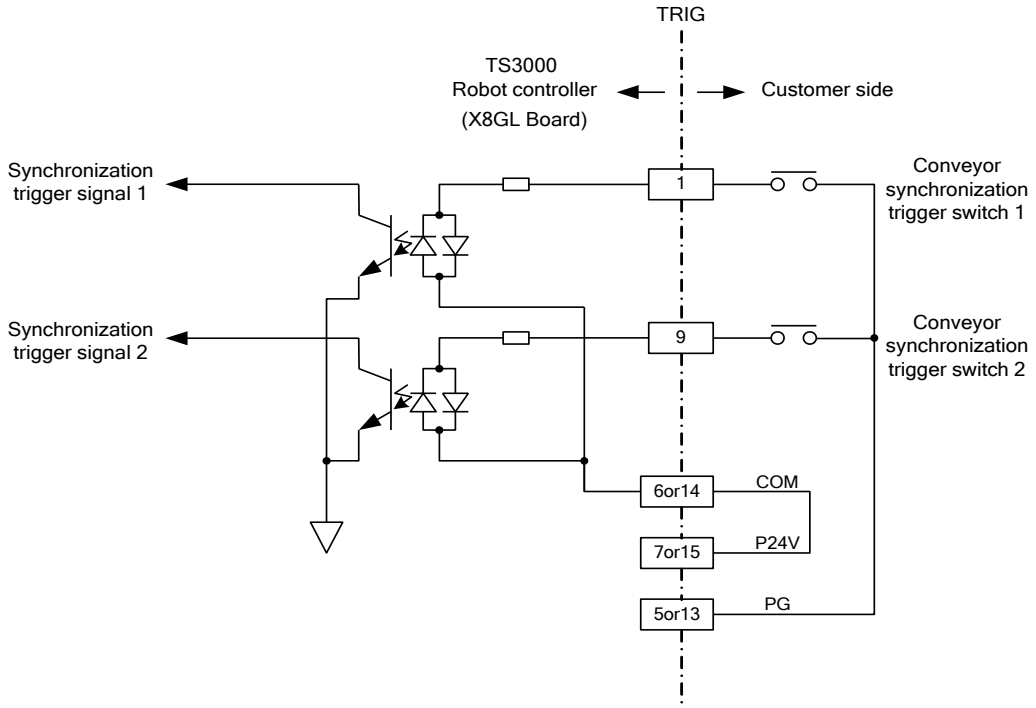



Figure 5.3 Connection of Conveyor Synchronization Start Trigger Switch (negative (-) common)




Figures 5.4 Connection of Conveyor Synchronization Start Trigger Switch (positive (+) common)

### 5.2.3. Specifications for Conveyor Trigger Input Signal Cable

Shown below are the specifications for the conveyor trigger input signal cable.

- Core wire: 0.2 - 0.52 $mm^2$ twisted wire
- Cable outside diameter: Max. 9 mm
- Shield: Integrated shield
- Cable length: 30 m or less

## 5.3. Connection of Conveyor Encoder

Both a signal from an incremental encoder and an input signal from a conveyor synchronization start trigger switch are necessary for using the conveyor synchronization function of the TS3000.

### 5.3.1. Specifications for Incremental Encoder

Use an encoder with 90-degree different A-phase and B-phase signals and an output circuit for RS-422 differential output (line driver).

Table 5.2 Encoder Specifications

| | |
|---|---|
| Max. response frequency of A/B phase | 250 kHz |
| Receiver IC | HD26LS32FP |
| Cable breakage detection | Yes |

### 5.3.2. Specifications for Incremental Encoder Connector

Connect the TS3000 and the incremental encoder to CONV*1.

Table 5.3 Connector Manufacture Name

| Connector Part | Type and Manufacturer |
|---|---|
| Connector | 54306-1419 (Molex) |
| Connector cover | 54331-0141 (Molex) |

For the connector signal array, see "5.3.3. Connection between Incremental Encoder and Robot Controller."

*1 Controller-side connector CONV: Type 52986-1479 (Molex)

### 5.3.3. Connection between Incremental Encoder and Robot Controller

As shown in Figures 5.5 and 5.6, use shielded twisted pair cables. Connect the encoders in pairs as shown in the diagrams. Note that a wrong connection lowers noise tolerance, causing malfunction. See Figure 5.5 for connection using the external power supply of the robot controller; see Figure 5.6 for connection using the internal power supply. (There are two channels for encoder input. Use them as needed.)

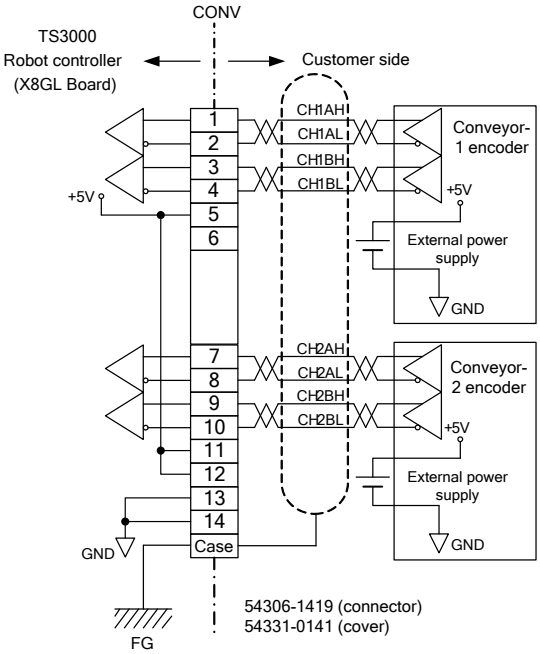



Figure 5.5 Connection between Incremental Encoder and Robot Controller (with external power supply used)

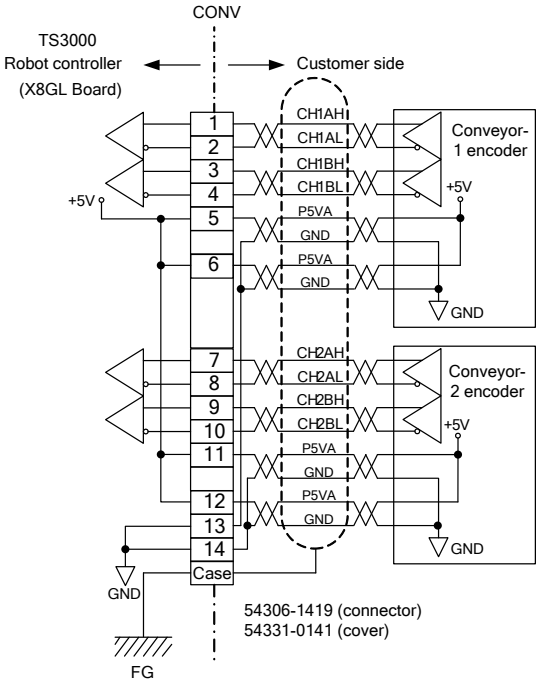



Figure 5.6 Connection between Incremental Encoder and Robot Controller (using with internal power supply)

### 5.3.4. Relationship between Encoder Cable Length and Number of Twisted-pair Power Cables

If the +5 V power from the robot controller is to be used as power supply to the encoder, connect the wires with consideration for the voltage drop in the cable.

**<Example of calculating the voltage drop in the cable>**

Assuming that the allowable voltage of the encoder is 4.5 V, the relationship between the cable length and the number of twisted-pair power cables is expressed in the following equation: (Calculated with the wire resistance as 110 Ω/km, the encoder consumption current as 150 mA, and the controller supply voltage as 5V)

$$V_e = 5 - \frac{110}{1000} \times l \times 2 \times \frac{150}{1000} \times \frac{1}{p}$$

Where:
$V_e$: Encoder voltage
$l$: Cable length
2: Number of reciprocations
$p$: Number of twisted-pair power cables
The following are obtained from the equation:

When the number of twisted-pair power cables is 1: Cable length 15 m or less
When the number of twisted-pair power cables is 2: Cable length 30 m or less
When the number of twisted-pair power cables is 3: Cable length 45 m or less

In the above conditions, it can be known that 4.5 V is obtained as the allowable voltage of the encoder. Therefore, **if a 45 m or longer cable is necessary, an external power supply is to be used.**

However, note that the calculation example above is applicable only in the above conditions. The customers are requested to calculate the voltage drop in the cable, referring to the calculation example above. Particularly note that the consumption current varies depending on your encoder. If an external power supply is used, a longer cable can be used, in which case the allowable maximum length is 100 m in terms of the signals.

### 5.3.5. Preparation of Cable

Prepare the cable according to the following:

a) Specifications for the incremental encoder cable

- Core wire: 0.2-0.52 $mm^2$ twisted wire
- Cable outside diameter: Max. $\phi$8.5 mm
- Shield: Integrated shield
- Cable length: See "5.3.4. Relationship between Encoder Cable Length and Number of Twisted-pair Power Cables."

Use only a cable conforming to the specifications. Do not use an unspecified cable. An unspecified wire can cause not only a short circuit or core wire melting due to overheating, but robot malfunction due to noise, too.

b) How to prepare the cable

The connector is of the solder cup type for connecting the wires by soldering. Prepare the cable, carefully connecting the wires to the connector and handling the shield according to Figure 5.7.

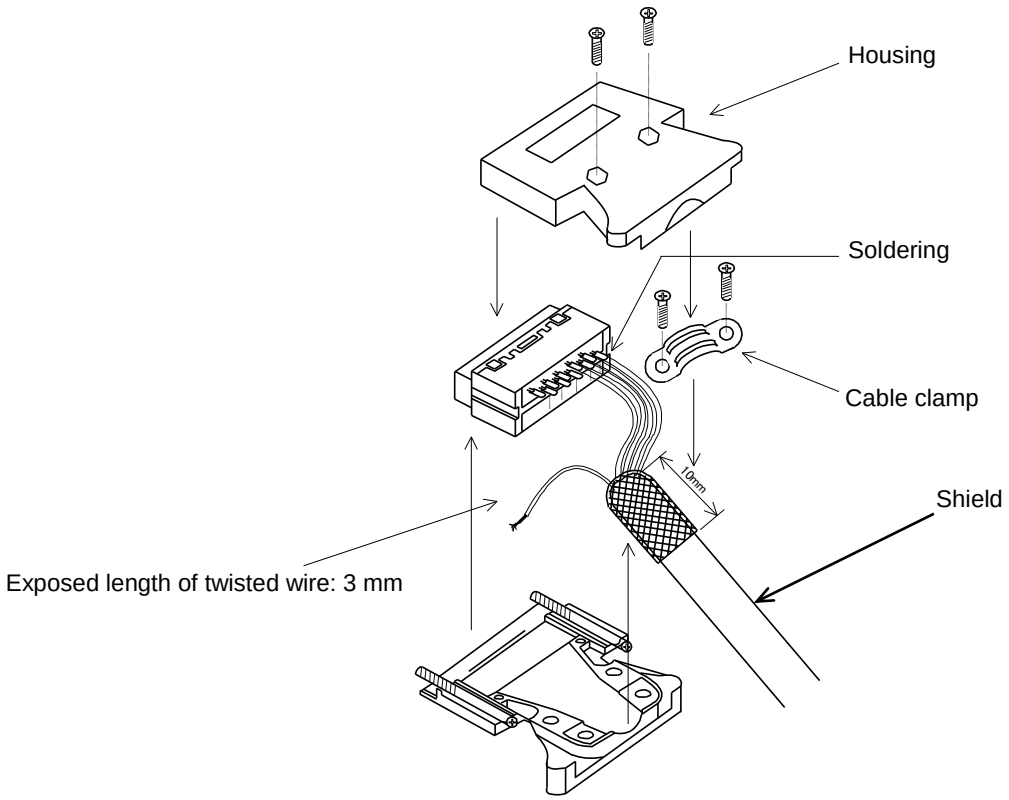



Figure 5.7 How to Prepare Cable

(Note) The cable shield is to be connected to the earth of the TS3000 robot controller through the housing. **Firmly connect the shield and the housing with the cable clamp.**

### 5.3.6. Determination of Adequate Encoder Resolution

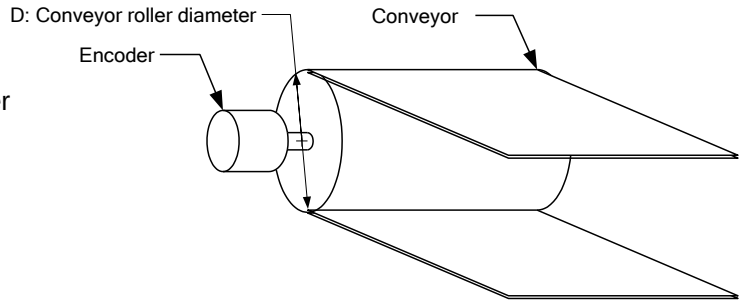
The minimum period of conveyor synchronization processing is 3 msec. If the number of conveyor encoder pulses that can be sampled every 3 msec is too low, accuracy to catch up with the conveyor speed lowers. Thus, select an encoder so that **"Vpulse: Number of sampled pulses per 3 msec" is <u>about 10 to 100 pulses</u>**.

"Vpulse: Number of sampled pulses per 3 msec" can be obtained from the following equation:

$$V_{pulse} = \frac{3}{1000} \cdot R_e \left( \frac{V_{CNV}}{\pi \cdot D} \cdot \frac{1}{G_m} \right) \qquad (1)$$

Where:

**<u>Re: Encoder resolution (ppr)</u>**
For example, if an encoder of 2000 (ppr) is used, it is processed as the quadruple resolution 8000 (ppr) by the TS3000 controller.




**<u>Vcnv: Conveyor speed (mm/sec)</u>**
Conveyor speed desired by the customer

Vcnv = 50 (mm/sec)

**<u>D: Conveyor roller diameter (mm)</u>**
Customer's conveyor roller diameter

D = 100 (mm)

**<u>Gm: Reduction gear ratio (number of conveyor roller revolutions/number of encoder revolutions)</u>**
The reduction gear ratio varies, depending on the customer's encoder mounting condition.

(Example) If the encoder revolves 25 times for a single revolution of the conveyor roller, Gm is as follows:

Gm* = 1/25 = 0.04

The following equation (2) can be obtained by modifying the equation (1) above:

$$R_e = V_{pulse} \cdot \frac{1000}{3} \cdot \frac{\pi \cdot D \cdot G_m}{V_{CNV}} \qquad (2)$$

Assigning the values of "Vcnv, Gm, D, and Vpulse" to the equation (2), the adequate encoder resolution "Re" turns out to be 840 to 8400 (ppr).

* Note Gm: Determine a design so that the reduction gear ratio will be a value divisible.

# Section 6 Programming and Teaching Operation

## 6.1. Notes on Programming

Note the following when creating conveyor synchronization programs.

### (1) Available motion instructions

During control for SYNC and UNSYNC instruction processing (i.e. during conveyor synchronization), **only the linear and circular interpolation (MOVES, MOVEC)** instructions can be specified as motion control instructions. If another motion control instruction is programmed, the system issues the error message "002-025: Invalid command" (invalid instruction execution error), with which the program stops.
However, once synchronization is started, the robot continues traveling parallel to the conveyor until it reaches the conveyor synchronization end position (<position-2;B> specified in the CONV instruction). When a motion control instruction in a subprogram in the library file (SCOL.LIB) is executed, the robot also continues traveling in the same manner until reaching the end position.

### (2) Positioning accuracy setting: ACCUR=COARSE

If a motion control instruction is specified in conveyor synchronization, specify ACCUR=COARSE as the positioning accuracy in the motion control instruction. This instruction must be specified before the SYNC instruction in the program. If **ACCUR=FINE (accuracy; initial state)** is specified as the positioning accuracy, the robot does not travel synchronously with the conveyor when the motion control instruction is executed.

### (3) Robot posture (CONFIG specification)

Specify the robot posture to be used during conveyor synchronization. The posture of the robot in traveling synchronously with the conveyor must be specified with CONFIG=RIGHTY (right shoulder system) or CONFIG=LEFTY (left shoulder system) before the SYNC instruction in the program. If a conflict is detected in the robot posture, the internally stored posture is forced to be used when the SYNC instruction is performed.

### (4) ON~DO~ instruction <condition-monitoring>

- If <condition-monitoring> needs to be monitored with the ON instruction during conveyor synchronization, execute the IGNORE instruction to cancel it immediately before the SYNC instruction, and then specify it again.
- As with the XIN instruction, specify the ON instruction as ON<condition-monitoring>DO GOTO<label> in the program. In this case, remember to move the robot to a safe place and end conveyor synchronization as needed.

### (5) PASS instruction (shortcut motion)

To use shortcut motion, specify ENABLE PASS immediately after the SYNC instruction, and DISABLE PASS immediately before the UNSYNC instruction. If shortcut motion is specified at different timing, it is discontinued or results in incorrect motion.

**(6) HERE instruction**

If a HERE instruction is specified in conveyor synchronization, the distance of conveyor travel is not included in the current position. In addition, the target position in the DEST instruction becomes indefinite. Caution should be taken when using a HERE instruction.

**(7) About travel distance during conveyor synchronization**

If synchronization between the points A and B "travel distance during conveyor synchronization: $|\overrightarrow{AB}|$" is complete (which is explained in the paragraph of the CONV instruction), the conveyor synchronization is automatically terminated irrespective of whether or not the "UNSYNC" instruction is specified. If an instruction which needs time to be processed is used, conveyor synchronization is continued during the processing time, so the robot eventually reaches the synchronization end position. (Examples of such an instruction: Program control instruction (e.g. WAIT~) and input/output control instruction (INPUT~, PRINT~)) **A program must be created so that the robot completes synchronous operation (e.g. picking up workpieces) between the points A and B "travel distance during conveyor synchronization: $|\overrightarrow{AB}|$."**

**(8) Program stop**

The program is stopped by performing a STOP instruction during conveyor synchronization. In this case, however, the conveyor synchronization is continued as explained above. If the program is stopped due to another error, the conveyor synchronization is also continued as long as the conveyor (encoder) is operating. For details, see "Section 7 Operation."

**(9) Breakage of conveyor encoder cable**

- If the encoder cable is broken, the system issues the error message "001-047 Conveyor Line Breakage." However, the robot does not stop, and it continues traveling synchronously with the conveyor **at the conveyor (encoder) speed effective immediately before the cable is broken**.

- If a breakage in the encoder cable is detected during conveyor synchronization, be sure to move the robot to a safe place. Otherwise, the robot can collide with the workpiece or the conveyor. To avoid the problem, execute the MOVES instruction to move the robot to a safe place, the UNSYNC instruction to end the conveyor synchronization, and then the STOP instruction to stop the program.

- The error "001-047 Conveyor Line Breakage" can be reset by only turning off the power. If the error occurs, turn off the power, identify the location of the cable breakage, repair the encoder cable, and then turn the power back on.

- Specify the XIN instruction immediately after the SYNC instruction, and the IGNORE instruction immediately before the UNSYNC instruction. If BREAK or PAUSE is specified during conveyor synchronization, it is treated as unspecified. Specify the XIN instruction as ON<condition-monitoring>DO GOTO<label> in the program.

### 6.2. Overview of Programming

This paragraph explains the programming of conveyor synchronization, showing the creation of a basic conveyor synchronization program.

① Specify the conveyor synchronization parameters **A, B, C, and D** in a CONV instruction.

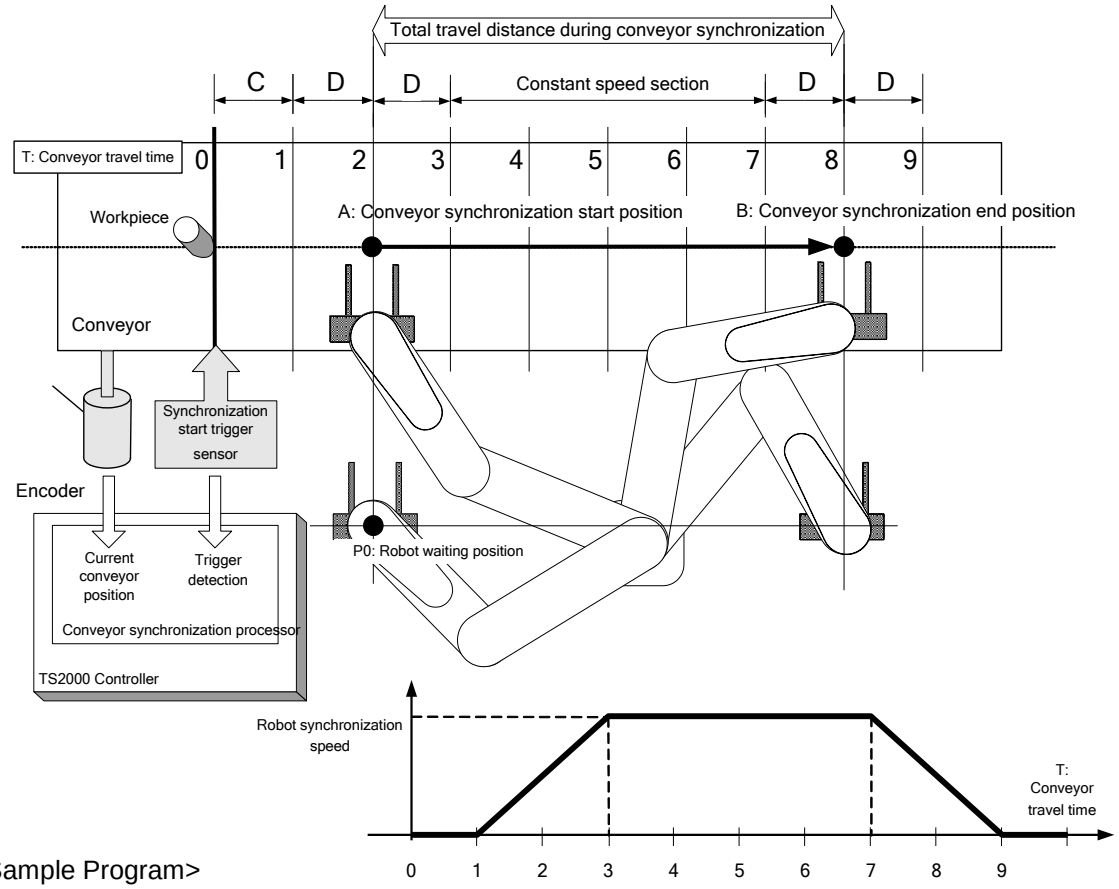
② Before the robot starts traveling synchronously **with the conveyor, it moves to the waiting position (=evacuation position) (P0)**.

③ Define a SYNC instruction to start conveyor synchronization processing at the same time as for trigger signal input.

④ Specify a motion **instruction to do operation** on the conveyor.

⑤ Define an **UNSYNC instruction** to complete conveyor synchronization processing.

⑥ Select the program and confirm that there are no syntax errors.




<Sample Program>

```
PROGRAM CONVTEST
        N = 1
        C = 10.0
        D = 5.0
        ACCUR = COARSE
        CONFIG = RIGHTY

        CONV( N, A, B, C, D )
        MOVE P0
        SYNC(N)

        WAIT CONVSTAT >= 4
```

Coarse is specified for the positioning accuracy.
The right shoulder system is specified for the robot posture.
**The parameters are defined.**
The robot moves to the waiting position.
**Synchronization is started when the LS is turned on.**
**The robot waits until the conveyor synchronization section changes to the "constant speed section."**

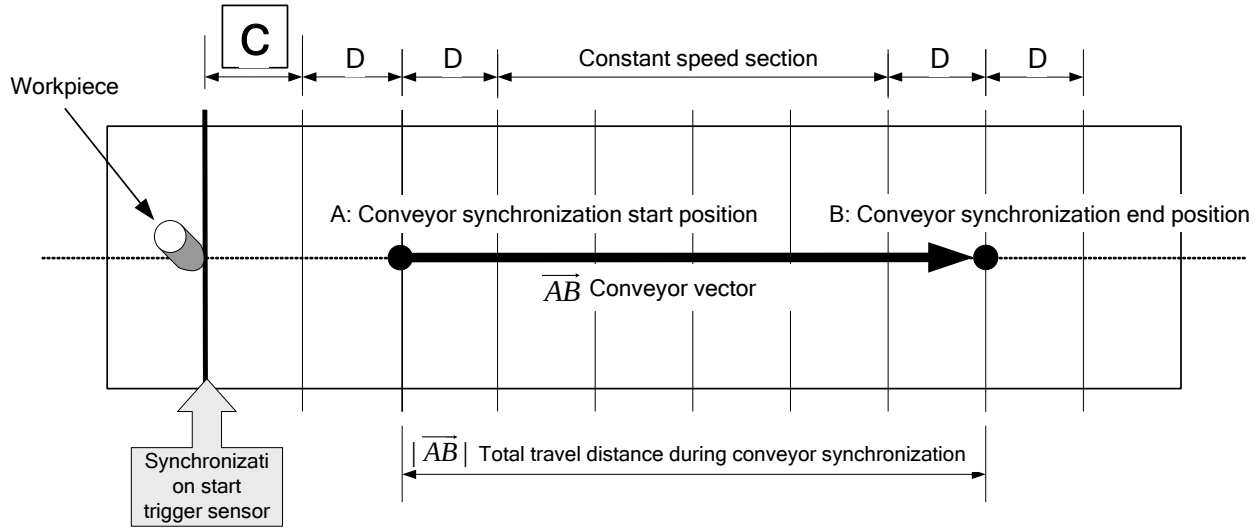
Actual operation program in conveyor synchronization

```
MOVES P1 + POINT( 0, 0, 100 )     The robot moves to the approach position.
DOUT(213,-214)                    The robot opens the hand.
MOVES P1 WITH SPEED = 20          The robot moves to the pick-up position.
DOUT(-213, 214)                   The robot closes the hand.
DELAY 0.3                         Robot motion is stopped for 0.5 sec.
MOVES P1+POINT( 0, 0, 100 )       The robot raises the hand.
MOVES P0                          The robot moves to the evacuation position.
UNSYNC(N)                         Synchronization is ended.
END
DATA
        A  =  ~     Conveyor synchronization start position
        B  =  ~     Conveyor synchronization end position
        P0 =  ~     Waiting/evacuation position
        P1 =  ~     Pick-up position
END
```

### 6.3. Setting of Conveyor Synchronization Parameters CONV (A,B,C,D)

This paragraph explains the setting/adjustment of the CONV instruction parameters "A,B,C,D" specified in the program.

#### 6.3.1. Determination of "C: Conveyor synchronization delay distance"




The C parameter specifies the distance between the actual trigger sensor position and the conveyor synchronization start position indicated by the point A (when 0 is specified in "D: Conveyor acceleration/deceleration distance" explained in the next paragraph).

The C parameter is useful if the trigger sensor is installed away from the conveyor synchronization start position for the sake of the line.
However, **the specification of a longer distance than necessary can cause an error in the distance itself. In order to avoid this problem, the system should be constructed so that the parameter can be specified with the shortest distance**.

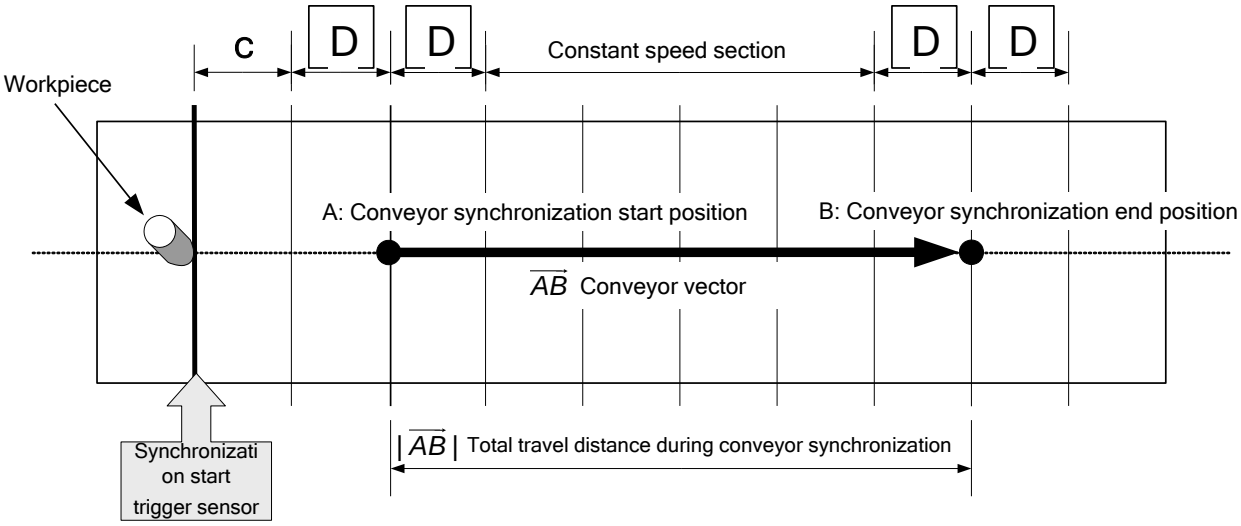
The C parameter can also be specified in a case other than the above. For example, if a **follow-up delay**[*1] occurs during synchronization, specify "C: Conveyor synchronization delay distance" to compensate for the delay.

Do not specify a negative value for **"C: Conveyor synchronization delay distance."** If a negative value is specified, the system issues the error message **"002-033: Conveyor Parameter Error."**

[*1] Probable causes of a follow-up delay are: (1) servo response delay, (2) delay of trigger signal input detection by the robot controller, and so on.

### 6.3.2. Determination of "D: Conveyor acceleration/deceleration distance"

The acceleration/deceleration distance D is specified for reducing shock to the robot and workpiece which occurs at the start of synchronous travel. It is specified, following the section C as shown below. (Do not specify the distance D before the section C.)




A target value (minimum) of the D value can be obtained from "Amax (mm/sec2): maximum acceleration for robot interpolation motion" in the following table and the synchronization speed.

| Type of Robot | Maximum Acceleration: Amax (mm/sec2) |
|---|---|
| TH450A TH550A TH650A | 10000 |
| TH850A TH1050A | 20000 |

<Calculation formula>

$$D_{min} >> V^2 / 2 \cdot A_{max}$$

The "V: Synchronization speed" is shown in the conveyor speed display "Veloc" of the conveyor utility function*1. However, this value is a minimum, and it needs to be increased depending on the robot hand and the mass of the workpiece being held by the robot. Use a value at least 2-3 times greater than the minimum value.

$$\boxed{D = 3 \cdot D_{min}}$$

If the robot seems to be affected by shock generated when it is accelerating to the conveyor speed in the adjustment stage explained later, readjust the value D.

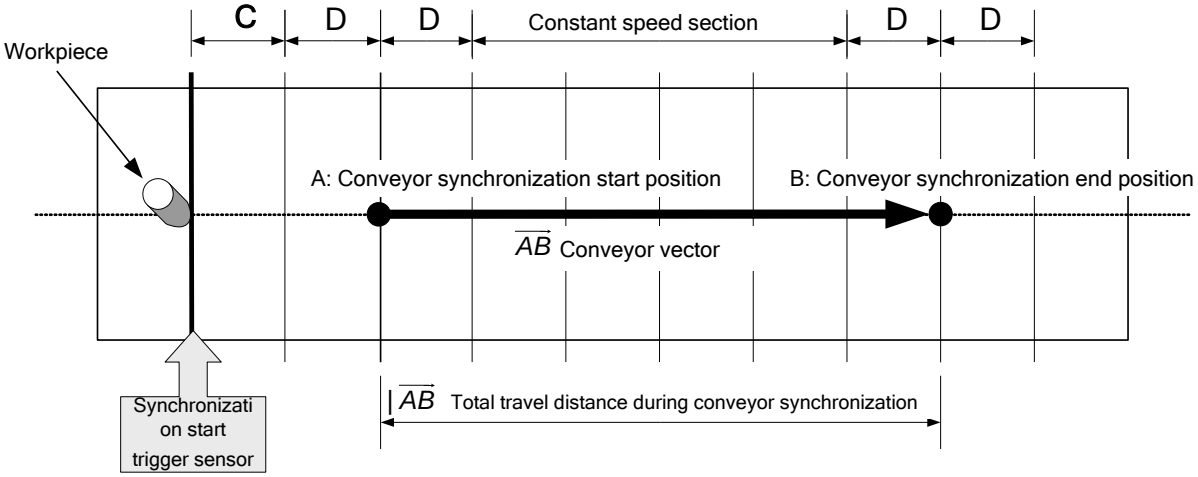
*1 For the conveyor utility function, see "7.4. Conveyor Synchronization Utility."

The actual operation section during conveyor synchronization is a zone where the robot synchronization speed is identical to the conveyor speed, that is to say, the constant speed section in the figure. The distance of the constant speed section is obtained from the following formula:

$$< \text{Distance of constant speed section} > = |\overrightarrow{AB}| - 2D$$

Therefore, if the distance D is longer than necessary, the actual operation section of the robot is decreased by the excess distance, so specify a proper value as the distance D.

### 6.3.3. Teaching of "A: Conveyor synchronization start position and B: Conveyor synchronization end position"




Assume that the vector connecting the points A and B is "conveyor vector $\overrightarrow{AB}$."

The distance between the two points is also called "Total travel distance during **conveyor synchronization:** $|\overrightarrow{AB}|$." This distance includes the acceleration/deceleration distance (D) necessary for conveyor synchronization. The robot performs synchronous operation in this section. However, the robot travels exactly at the conveyor speed only in the "constant speed section" in the diagram.

Teach the robot the two points A and B in a way that the conveyor vector $\overrightarrow{AB}$ will be parallel to the conveyor.

Teach the robot that the point A is located the distance "C+D" from the trigger sensor position to the conveyor vector direction.*1

*1 An error in the distance to the conveyor vector direction that is taught here can be absorbed by adjusting the conveyor synchronization delay distance (C).

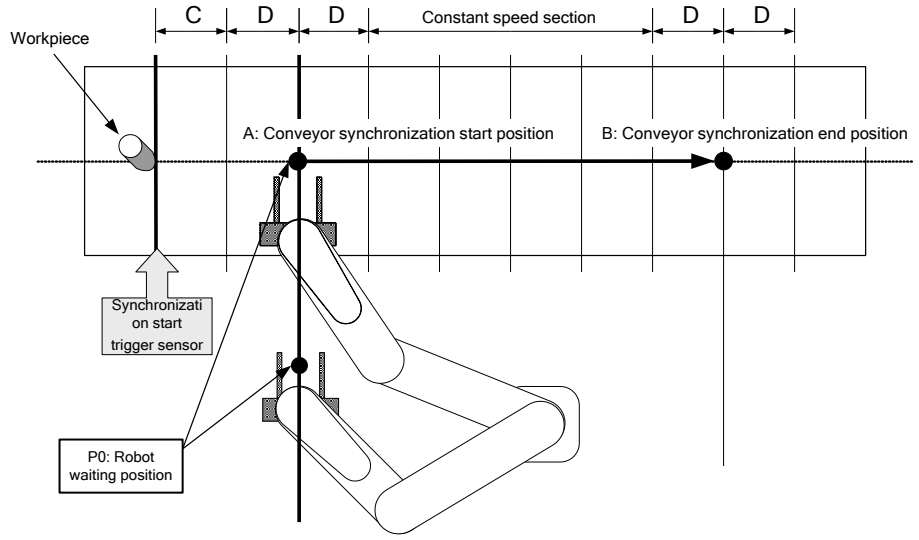
### 6.4. Teaching Operation

This paragraph explains the "teaching operation" of the conveyor synchronization function.

**CAUTION**

- Before teaching operation can be performed, the conveyor must be stopped. If the synchronization start trigger sensor is activated during conveyor operation, an unexpected motion or collision can occur depending on the teaching operation condition. (The robot at the waiting position stays still only until the trigger sensor is turned on, but it is operable at any time.)
- Define a safe place as a robot's evacuation position where the robot will never be involved in any collision with workpieces on the moving conveyor even if robot motion is stopped. (The robot waiting position is desired to be the evacuation position.)
- Adjust the trigger sensor so that it will be always turned on at a fixed position without varying the detection position of objects (e.g. workpieces). If the sensor is turned on due to a mixture of different objects or a non-adjusted position, a follow-up delay or collision will occur.
- In test run mode, do not execute a program (between the SYNC and UNSYNC instructions) in steps and do not use the machine lock (dry run) or MODE: MOTION. Otherwise, an unexpected collision can occur if the conveyor is operating.

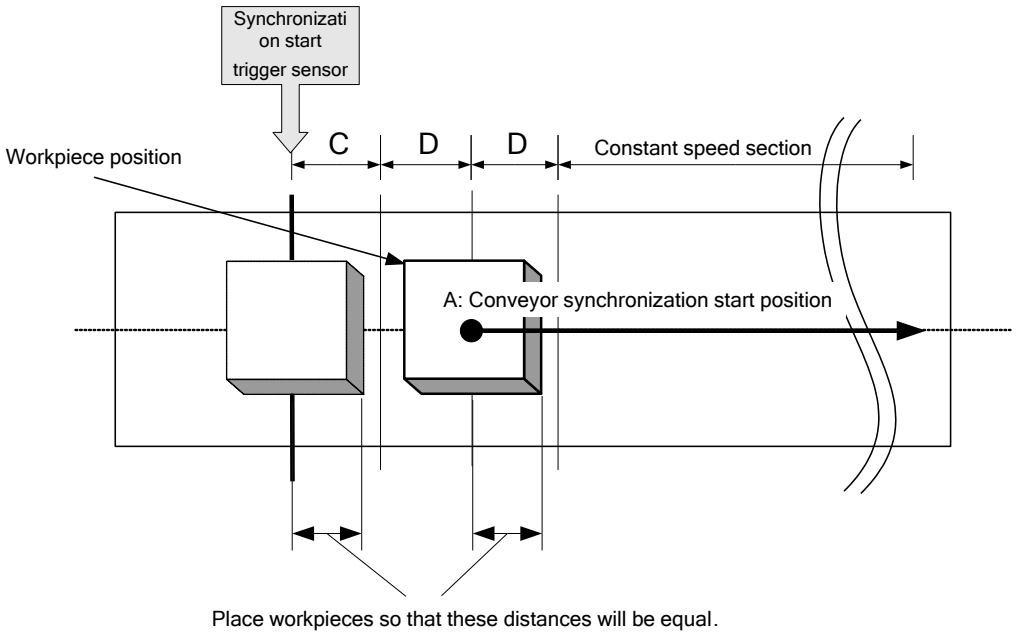
#### 6.4.1. Teaching of Robot Waiting Position




Teach the robot waiting position "P0" to the robot.

### 6.4.2. Teaching of Motion Positions in Conveyor Synchronization

Teach the robot the motion positions to be used with motion instructions in conveyor synchronization. Actual workpieces must be placed on the conveyor when teaching the robot how to handle the workpieces. At this time, pay attention to the relationship between the "trigger sensor position" and the "workpiece detection position."




If the total of the conveyor speed and the motion speed specified in the motion instruction exceeds 1000 (mm/sec), the robot can be stopped due to the error "Axis1-4 Exceed Speed." If an instruction to specify a motion to the direction of the conveyor vector $\overrightarrow{AB}$ is used, the program must be created with attention to the motion speed.

### 6.5. Readjustment/Confirmation of Conveyor Synchronization Parameters "CONV (A,B,C,D)"

Check the conveyor synchronization parameters. In this paragraph, check if the actual operation program in conveyor synchronization is completed in the constant speed section. For the program to be completed in the section, the following formula must be satisfied:

$|\overrightarrow{AB}| - 2 \cdot D$ >> Lcv(mm): Conveyor travel distance during actual operation

If the synchronization is not completed, make the following adjustments as needed:

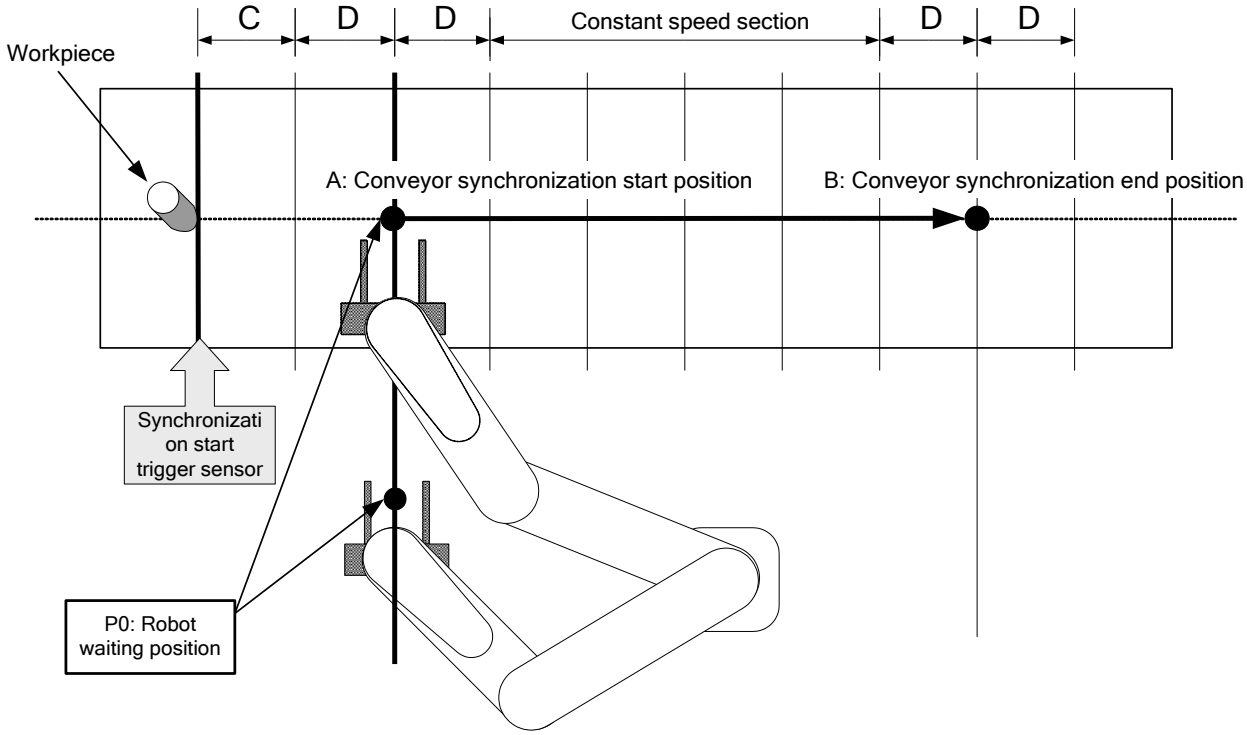
(A) Increase $|\overrightarrow{AB}|$.

(B) Make the distances C and D as short as possible.

(C) Reduce the processing time of the "actual operation program."

(D) Decrease the conveyor speed.

(E) Change the trigger sensor position.



**Step ①**

Measure the processing time (Tcv: sec) for the **actual operation program in conveyor synchronization** that is defined between the SYNC and UNSYNC instructions. To do so, change the program as shown below.

```
PROGRAM CONVTEST
        :
        CONV( N, A, B, C, D )
        MOVE P0
        ' SYNC(N)                  → Comment out
        TIMER = 100.0
        'WAIT CONVSTAT >= 4        → Comment out
        MOVES P1 + POINT( 0, 0, 100 )
        DOUT(213,-214)
MOVES P1 WITH SPEED = 20
        DOUT(-213, 214)
        DELAY 0.3
        MOVES P1+POINT( 0, 0, 100 )
        MOVES P0
        PRINT "Tcv=", TIMER–100.0, CR   → The processing time (sec) is shown on
                                          the teach pendant display.
        ' UNSYNC(N)               → Comment out
END
```

Actual operation program in conveyor synchronization
Tcv (sec): Actual operation time
Lcv (mm): Conveyor travel distance during actual operation

Note 1: The conveyor synchronization start instruction is commented out in the above program, so **conveyor synchronization is not performed**.

Note 2: Before the above program can be run, the conveyor must be stopped after the removal of the workpieces from the conveyor.

**Step ②**

After checking the operation of the program in test run mode (MODE: MANU), run the program in internal auto mode (MODE: INT) to measure the processing time. The measured time is shown with "Tcv (sec)" on the teach pendant (hereinafter called TP) display when the program is completed.

**Step ③**

Subsequently, move the robot to a safe place, and start the conveyor.

Measure the conveyor speed (Vcv: mm/sec) using the conveyor utility function*1. The conveyor travel distance (Lcv: mm) during actual operation is obtained by assigning the values of Tcv and Vcv to the following formula. If the obtained distance is significantly smaller than the distance of the constant speed section, in other words, if the following formula is satisfied, the setting and checking of the CONV (A,B,C,D) parameters are complete.

$$|\overrightarrow{AB}| - 2 \cdot D \quad >> \quad \text{Lcv(mm)} \quad = \quad \text{Vcv} \times \text{Tcv}$$

*1 See "7.4. Conveyor Synchronization Utility," for how to use the conveyor utility function.

**Step ④**

Check the follow-up performance for the conveyor speed.
In this step, while workpieces are on the moving conveyor, get the robot to start travelling synchronously with the conveyor when the trigger signal is turned on. However, do the following to enable conveyor synchronization:

(A) Remove the workpieces.

(B) Enable SYNC and UNSYNC.

(C) Comment out all the lines of the actual operation program.

(D) If the waiting position differs from the point A (conveyor synchronization start position) and the distance between the conveyor and robot is large, it is difficult to check follow-up performance. In this case, change the waiting position (P0) to the point A (conveyor synchronization start position).

```
PROGRAM CONVTEST
        :
        CONV( N, A, B, C, D )
        MOVE P0
        MOVE   A          →     The waiting position is changed to the point A.
        SYNC(N)
        WAIT CONVSTAT >= 4
        'MOVES P1 + POINT( 0, 0, 100 )
        'DOUT(213,-214)
        'MOVES P1 WITH SPEED = 20
        'DOUT(-213, 214)
        'DELAY 0.3
        'MOVES P1+POINT( 0, 0, 100 )
        'MOVES P0
        UNSYNC(N)
END
```

Actual operation program in conveyor synchronization
→ Comment out all the lines.

Check the following items while repeatedly moving the robot synchronously with the conveyor by running the above program:

**a. Presence/absence of shock to the arm when conveyor synchronization is started**

Readjust "D: Conveyor acceleration/deceleration distance" according to Paragraph 6.3. if necessary.

**b. Robot waiting position**

At the point immediately after the start of synchronization:
If the robot gets behind workpieces: Decrease the distance C.
If the robot goes ahead of workpieces: Increase the distance C.

**c. Travel value*1 per pulse of encoder specified in the "CONVEYOR.PAR" file**

If the robot starts traveling slower than the conveyor in the latter half of the travel distance during conveyor synchronization: Increase the travel distance per pulse.
If the robot starts traveling faster than the conveyor in the latter half of the travel distance during conveyor synchronization: Decrease the travel distance per pulse.

This completes the readjustment/confirmation of the conveyor synchronization parameters "CONV (A,B,C,D)."

*1 See "3.6. ⑧ Travel value per pulse of encoder."

### 6.6. Program Operation Check

After checking that the robot can handle workpieces on the moving conveyor at the proper timing, carefully change the program back to the original, and perform actual operations such as having the robot pick up workpieces on the moving conveyor.

**⚠ CAUTION**

- Thoroughly read "Section 7 Operation" before attempting to start the program. Particularly if having performed a stop operation during conveyor synchronization, follow the recovery procedures in "7.1. Stop Operation" and "7.2. Reset Operation."
- In test run mode, the limit speed (LSPEED) is restricted to 25%. Thus, it is dangerous to perform operation in this condition because synchronous operation (e.g. picking up workpieces) is not completed in the range of travel distance during conveyor synchronization. Do not perform a test run after completing the operation procedures in the previous paragraphs.
- Check the program operation in internal auto mode without changing LSPEED unnecessarily.
- If operation in test run mode is necessary later, stop the conveyor, and make sure that there are no obstacles in the robot's working envelope.

This completes the operation procedures. If a fault is detected, check the operations with the procedures explained so far in Section 6.

<Test Program for Final Check>

```
PROGRAM CONVTEST
        N = 1
        C = 10.0
        D = 5.0
        ACCUR = COARSE                  Coarse is specified for the positioning accuracy.
        CONFIG = RIGHTY                 The right shoulder system is specified for the robot posture.
        CONV( N, A, B, C, D )           The parameters are defined.
        MOVE P0                         The robot moves to the waiting position.
        SYNC(N)                         Synchronization is started when the LS is turned on.
          WAIT CONVSTAT >= 4            The robot waits until the conveyor synchronization
                                        section changes to the "constant speed section."
         MOVES P1 + POINT( 0, 0, 100 )  The robot moves to the approach position.
          DOUT(213,-214)                The robot opens the hand.
         MOVES P1 WITH SPEED = 20       The robot moves to the pick-up position.
          DOUT(-213, 214)               The robot closes the hand.
          DELAY 0.3                     Robot motion is stopped for 0.5 sec.
          MOVES P1+POINT( 0, 0, 100 )   The robot raises the hand.
          MOVES P0                      The robot moves to the evacuation position.
        UNSYNC(N)                       Synchronization is ended.
END
        DATA
                A  =  ~     Conveyor synchronization start position
                B  =  ~     Conveyor synchronization end position
                P0 =  ~     Waiting/evacuation position
                P1 =  ~     Pick-up position
        END
```

### 6.7. Sample Programs

This paragraph shows sample programs using conveyor synchronization control instructions. If the programs are to be applied to your robot, review the programs in accordance with the working envelope and environment of the robot.

#### 6.7.1. Handling Operation

```
PROGRAM HANDLEING
      ACCUR=COARSE                Coarse is specified for the positioning accuracy.
      CONFIG=RIGHTY               The right shoulder system is specified for the robot
                                  posture.
      N=1
      C=10.0
      D=5.0
      CONV( N, A, B, C, D )       The parameters are defined.
      MOVE P0                     The robot moves to the waiting position.
      SYNC(N)                     Synchronization is started when the trigger signal is
                                  turned on.
        MOVES P1+POINT( 0, 0, 100 )  The robot moves to the approach position.
        DOUT(213,-214)            The robot opens the hand.
        MOVES P1 WITH SPEED = 20  The robot moves to the pick-up position.
        DOUT(-213,214)            The robot closes the hand.
        DELAY 0.3                 Robot motion is stopped for 0.5 sec.
        MOVES P1+POINT( 0, 0, 100 )  The robot raises the hand.
        MOVES P0                  The robot moves to the evacuation position.
      UNSYNC(N)                   Synchronization is ended.
END
DATA
      A  =  ~                     Conveyor synchronization start position
      B  =  ~                     Conveyor synchronization end position
      P0 =  ~                     Waiting/evacuation position
      P1 =  ~                     Pick-up position
END
```

- Be sure to specify the positioning accuracy and robot posture.
- A variable is used as each parameter.
- For the robot waiting position, see "Section 1 Overview."
- The hand is opened and closed with the auxiliary output signal. (DOUT instruction used)
- The Z-axis (vertical) is used, so the robot motion speed is kept low. (WITH clause used)
- The evacuation position at this time is shifted to the distance of synchronous travel from the original point P1 to the conveyor traveling direction.

### 6.7.2. Pick & Place Operation

```
PROGRAM PICKUP
        ACCUR=COARSE            Coarse is specified for the positioning accuracy.
        N=2
        CONV(N,A2,B2,15.0,10.0) The conveyor-2 parameters are defined.
        ON XIN(2) DO GOTO L200  The encoder cable 2 is checked for breakage.
        N=1
        CONV(N,A1,B1,40.0,10.0) The conveyor-1 parameters are defined.
        ON XIN(1) DO GOTO L200  The encoder cable 1 is checked for breakage.
        ……..
        ……..
L100:
        CONFIG=RIGHTY           The right shoulder system is specified for the robot
                                posture.
        MOVE P1                 The robot moves to the waiting position 1.
        SYNC(1)                 Synchronization is started when the LS1 is turned on.
        ……..                    The robot picks up workpieces on the conveyor 1.
        MOVES P1                The robot moves to the evacuation position 1.
        UNSYNC(1)               Synchronization is ended.
        IGNORE XIN(1)
        ……..
        CONFIG=LEFTY            The left shoulder system is specified for the robot
                                posture.
        MOVE P2                 The robot moves to the waiting position 2.
        SYNC(2)                 Synchronization is started when the LS2 is turned on.
        ……..                    The robot picks up workpieces on the conveyor 2.
        MOVES P2                The robot moves to the evacuation position 2.
        UNSYNC(2)               Synchronization is ended.
        IGNORE XIN(2)
        ……..
        GOTO L100
L200:
        PRINT “C/N ERROR=”,N,CR An error message is shown on the TP display.
        STOP                    The program is stopped.
END
DATA
        A1 =~                   Synchronization start position for conveyor 1
        B1 =~                   Synchronization end position for conveyor 1
        P1 =~                   Robot waiting/evacuation position for conveyor 1
        ……..                    …….. for conveyor 1
        A2 =~                   Synchronization start position for conveyor 2
        B2 =~                   Synchronization end position for conveyor 2
        P2 =~                   Robot waiting/evacuation position for conveyor 2
        ……..                    …….. for conveyor 2
END
```

- The conveyor 1 is for taking out workpieces; the conveyor 2 is for unloading workpieces.
- Each encoder cable is checked for breakage immediately after the parameters are defined.

# Section 7 Operation

This section explains how to operate the conveyor synchronization function.

## 7.1. Stop Operation

The following stop operations are available with the TS3000 controller:

- Stop using STOP or external stop signal input
- BREAK
- Stop using selection of master mode switch
- Stop using SCOL-language "STOP instruction"
- Stop using MODE: STEP or MOTION
- FEED HOLD

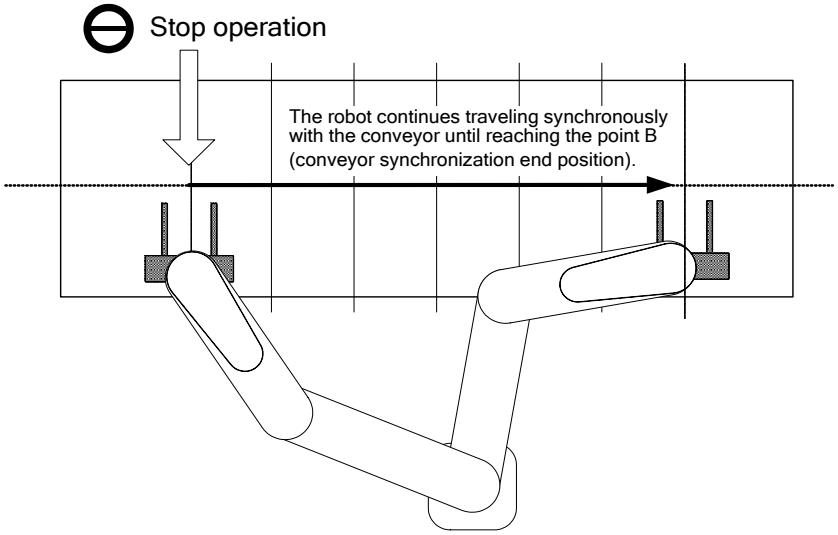
→ A. Stop with servo ON

- Emergency stop
- Servo OFF (external servo-OFF signal input)

→ B. Stop with servo OFF

If a stop operation is performed during conveyor synchronization (i.e. when the robot is traveling synchronously with the conveyor):

**A. Stop with servo ON:**
If a stop operation is performed during conveyor synchronization, the program is stopped, but the robot continues traveling synchronously with the conveyor until reaching the conveyor synchronization end position (B).




If the robot is left in this condition, it is to stop when reaching the point B (conveyor synchronization end position). As the result, the robot hand interferes with a workpiece on the conveyor in synchronization, which can lead to a collision accident. To prevent the above accident, construct the system keeping the following in mind:

① If an external device (including a conveyor) is stopped in an emergency, the robot shall also be stopped in an emergency.

② If the robot is stopped in an emergency, the external device shall also be stopped in an emergency.

③ If the robot program is stopped for some reason (e.g. due to an error or a stop operation performed by the user), the conveyor shall also be stopped.

**B. Stop with servo OFF**
The robot is stopped instantaneously; synchronization processing is ended instantaneously.

In consideration of the above, keep the following in mind when performing a stop operation during conveyor synchronization.

## ⚠ CAUTION

- Even if the program is stopped during conveyor synchronization, the robot can be continuously operated by immediately restarting the program. In principle, however, after operator intervention to perform a stop operation, be sure to perform a reset operation, move the robot to a safe place in manual guide mode, and then restart robot operation after resetting the program. (For details, see "7.2. Reset Operation.")

- Unless an abnormal state is detected, do not perform any "stop operation with servo OFF" or "emergency stop operation" while the conveyor synchronization function is performed. Otherwise, the robot hand interferes with a workpiece on the conveyor in synchronization, which can lead to a collision accident.

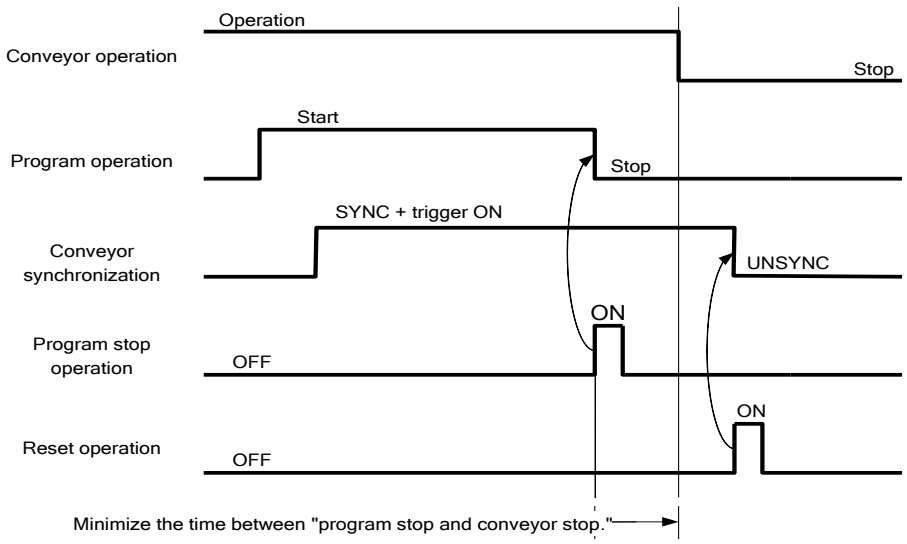
- To prevent the above accident, construct the system keeping the following in mind:
  ① If an external device (including a conveyor) is stopped in an emergency, the robot shall also be stopped in an emergency.
  ② If the robot is stopped in an emergency, the external device shall also be stopped in an emergency.
  ③ If the robot program is stopped for some reason (e.g. due to an error or a stop operation performed by the user), the conveyor shall also be stopped.

### 7.2. Reset Operation

If conveyor synchronization is stopped before completion, the conveyor synchronization processing is stopped by performing a reset operation. In other words, the UNSYNC instruction (synchronization end instruction) is executed by performing the reset operation.
There are three types of reset operation.

① Program reset
② Step reset
③ Cycle reset

If a stop operation in Paragraph 7.1 is performed during conveyor synchronization, be sure to perform a "reset operation" before restarting the program. The reset operation must be performed exactly according to the following procedure. Otherwise, **the robot hand interferes with a workpiece on the conveyor in synchronization, which can lead to a collision accident.**




Note 1: After performing a stop operation, stop the conveyor before performing a reset operation.
Construct such a system that the time between "program stop and conveyor stop" in the figure will be minimized.
Note 2: If a stop operation explained in **"B. Stop with servo OFF"** of Paragraph 7.1 is performed, specify a nearly zero value as the time between "program stop and conveyor stop" in the figure.

### 7.3. Conveyor Stop

The system issues the error message "001-048: Conveyor Stop" in the following case:
- conveyor synchronization is started when the synchronization trigger signal is turned on, but
- the number of pulses from the encoder is zero (0) for the specified time in the construct parameter.

The error is a level-1 error, so the program runs continuously without being stopped.

Conveyor (encoder) speed variation should be within ±7.5% of the average uniform velocity (speed constant) of your conveyor. Otherwise, workpieces and robot hand might be damaged in addition to a failure in obtaining correct synchronization accuracy.

### 7.4. Conveyor Synchronization Utility (Utility → Conv)

This paragraph explains the utility function for monitoring the conveyor synchronization function.

```
CONV UTILITY      <CH1>      <CH2>
 Conv Form    :    LINEAR     ARC(+)
 Conv Veloc   : -****.***  -****.****
 Conv Status  :          0          0
 Enc Pulse    :      32767      32767
 Mes Pulse    :     632417          0
 Mes Trip     : -****.***  -****.****
 MES1  MES2
```

① Conveyor trigger signal monitoring: When the conveyor trigger signal is turned on, <CH1> and <CH2> are "highlighted."

<CH1>: Trigger signal OFF
<CH1>: Trigger signal ON

② Conv Form: Indicates the type of conveyor synchronization.

LINEAR: Linear conveyor

ARC(+): Arc conveyor (clockwise)
ARC(-): Arc conveyor (counterclockwise)

③ Conv Veloc: Indicates the conveyor speed.

④ Conv State: Indicates the conveyor status.

0: Asynchronous state
1: Synchronization signal wait state
2: Traveling in delay section
3: Traveling in acceleration section
4: Traveling in constant speed section
5: Traveling in deceleration section
6: Synchronization end processing
-1: Conveyor cable breakage

⑤ Enc Pulse: Pulse counter value of the conveyor encoder

⑥ Mes Pulse: Total number of counted pulses of the conveyor encoder

To start counting the total number of encoder pulses, press the function key <F1: MES1> or <F2: MES2>. For the counting method, see "7.4.1. Counting of Total Number of Encoder Pulses."

⑦ Mes Trip: Measured distance of conveyor travel

The value displayed is the result of multiplying the value of "Mes Pulse" by the "travel value per pulse of encoder" in the [C00] CONVEYOR PARAMETER" field.

### 7.5. Counting of Total Number of Encoder Pulses

① Using the cursor, select and press the function key <F1: MES1> or <F2: MES2> corresponding to the ch of which pulses are counted.

```
CONV UTILITY      <CH1>        <CH2>
 Conv Form    :    LINEAR       ARC(+)
 Conv Veloc   : -****.***   -****.****
 Conv Status  :           0            0
 Enc Pulse    :       32767        32767
 Mes Pulse    :      632417            0
 Mes Trip     : -****.***   -****.****
 MES1   MES2
```



Select MES1 when counting the number of pulses of Ch1.

② Select "MES1" to count the pulses of the Ch1; select "MES2" to count the pulses of the Ch2. Move the conveyor the distance ΔL, and then press <SET>.

```
CONV UTILITY      <CH1>        <CH2>
 Conv Form    :    LINEAR       ARC(+)
 Conv Veloc   : -****.***   -****.****
 Conv Status  :           0            0
 Enc Pulse    :       32767        32767
 Mes Pulse    :      632417            0
 Mes Trip     : -****.***   -****.****
 SET
```



③ When <SET> is pressed, the following are displayed:

- Mes Pulse: Total number of counted pulses
- Mes Trip: Travel distance obtained by converting the unit of Mes Pulse

To count pulses again, press MES1 or MES2, and perform the operations in ① and ②.

```
CONV UTILITY      <CH1>        <CH2>
 Conv Form    :    LINEAR       ARC(+)
 Conv Veloc   : -****.***   -****.****
 Conv Status  :           0            0
 Enc Pulse    :       32767        32767
 Mes Pulse    :      632417            0
 Mes Trip     : -****.***   -****.****
 MES1   MES2
```

(Note)
"Mes Pulse" can be displayed normally only when the conveyor parameter is specified. How to specify the parameter: Calculate a travel distance per pulse from "Mes Pulse" (number of counted pulses) and the distance of conveyor travel (measured value), and specify the calculation result in the [C00] CONVEYOR PARAMETER" field. For details, see "3.6. ⑧ Travel value per pulse of encoder."

# Section 8 Troubleshooting

## 8.1. Error Messages

If an error is detected by the self-diagnosis function, the corresponding error message is shown on the teach pendant display. Identify the cause of the error according to the following table, and take correct action.

| Error Code | Error | Cause |
|---|---|---|
| 001-047 | Conveyor Line Breakage | "Encoder breakage during conveyor synchronization"<br>• The encoder cable of the conveyor with the specified number is broken or the encoder is faulty. Check the wiring.<br>• Even if this error occurs, the robot does not stop, and it continues traveling synchronously with the conveyor at the conveyor (encoder) speed effective immediately before the cable is broken.<br>• If a breakage in the encoder cable is detected during conveyor synchronization, be sure to move the robot to a safe place. Otherwise, the robot can collide with the workpiece or the conveyor.<br>• This error can be reset by only turning off the power. Thus, if the error occurs, turn off the power, identify the location of the cable breakage, repair the encoder cable, and then turn the power back on. |
| 001-048 | Conveyor Stop | "Conveyor in stop state"<br>A period of time during which the conveyor is in stop state has reached the stop judgment time specified in the parameter.<br>• The conveyor was in stop state when UNSYNC is executed. |
| 002-033 | Conveyor Parameter Error | "CONV parameter error"<br>• The parameter value is outside the allowable range. |

### 8.2. Troubleshooting

This paragraph lists error messages concerning conveyor synchronization processing. If the step number of an error location indicates a CONV, SYNC, or UNSYNC instruction, see the following table to identify the cause of the error, and take correct action.

| Instruction | Error Message and Cause |
|---|---|
| CONV | "002-042:Data Type error" Abnormal type error<br>• The type of a variable as an argument is wrong. |
| | "002-025:Invalid command" Invalid effective instruction error<br>• The conveyor synchronization function is not enabled. |
| | "002-033:Conveyor Parameter Error" CONV Parameter abnormal error<br>• The specified value is outside the allowable range. |
| SYNC | "002-041:Invalid Variable" Invalid effective instruction error<br>• The specified value is outside the allowable range. |
| | "002-025:Invalid command" Invalid effective instruction error<br>• The conveyor parameter for <number> is not defined.<br>• The programming procedure is wrong. |
| UNSYNC | "002-041:Invalid Variable" Invalid effective instruction error<br>• The specified number is outside the allowable range. |
| | "002-025:Invalid command" Invalid effective instruction error<br>• The conveyor parameter for <number> is not defined.<br>• Conveyor synchronization is not started yet. |
| XIN | "002-042:Data Type error" Abnormal type error<br>"002-025:Invalid command" Invalid effective instruction error<br>• The specified number is outside the allowable range. |
| Other | "002-025:Invalid command" Invalid effective instruction error<br>• An invalid motion control instruction is specified between the SYNC and UNSYNC instructions. |